

Restrictions on Conditional Correlations in SVARs^{*}

Myunghyun Kim[†] Sunho Lee[‡]

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Abstract

This paper introduces the *correlation restriction* as an identifying restriction in Bayesian structural vector autoregressions (SVARs). It is based on the economic distinction that demand shocks move output and prices in the same direction, whereas supply shocks move them in opposite directions. The correlation restriction ensures that the structural parameters generate a positive (negative) conditional correlation between the two variables for demand (supply) shocks. We use the restriction to identify monetary policy shocks (demand shocks) and oil supply shocks (supply shocks) and show that it is very useful in identifying the two shocks. In the case of monetary policy shocks, the correlation restriction narrows the credible sets for the impulse response functions (IRFs) and helps resolve the price puzzle. Moreover, the correlation restriction helps to sharply identify oil supply shocks by narrowing the credible sets for output and prices, providing clearer evidence of an output decline and a price increase.

JEL Classification: C32, E31, E52, Q43

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[†] Department of Economics, Sungkyunkwan University, 25-2, Sungkyunkwan-ro, Jongno-gu, Seoul, 03063, Korea (e-mail: mhkim7812@gmail.com).

[‡] Office of Economic Modeling & Policy Analysis, Bank of Korea, 39, Namdaemun-ro, Jung-gu, Seoul, 04531, Korea (e-mail: econ.preference@gmail.com).

1 Introduction

The key features of demand and supply shocks are that a positive demand shock, such as an expansionary monetary policy shock, increases output and prices, whereas a supply shock, such as an oil supply shock reducing world oil production, decreases output and increases prices. This distinction motivates imposing a positive conditional correlation between output and prices for demand shocks and a negative one for supply shocks.¹ Although demand and supply shocks are easily distinguishable, and their key features are well understood in economic theory, these features are not typically imposed directly as identifying restrictions in Bayesian structural vector autoregressions (SVARs). Therefore, this paper incorporates the key features of demand and supply shocks into Bayesian SVARs through the *correlation restriction* and demonstrates its effectiveness in precisely identifying monetary policy shocks (demand shocks) and oil supply shocks (supply shocks).

The correlation restriction ensures that the structural parameters produce a positive (negative) conditional correlation between output and prices for demand (supply) shocks. Specifically, a correlation restriction for demand shocks rules out the structural parameters that generate negative conditional correlations between output and prices for demand shocks. In contrast, a correlation restriction for supply shocks rules out the structural parameters that produce positive conditional correlations between the two variables for supply shocks. The correlation restriction is highly flexible, and thus it can be easily applied to other restrictions. Despite its flexibility, our study focuses on combining the correlation restriction and sign restrictions on the structural parameters or the impulse response functions (IRFs) in SVARs.²

Since Uhlig (2005), it has become popular to use minimal, uncontroversial sign restrictions on the IRFs or the structural parameters when identifying structural shocks (e.g., monetary policy shocks and oil supply shocks), as they are generally thought to be weaker than classic restrictions, such as the recursive restrictions. Moreover, according to Rubio-Ramírez, Waggoner, and Zha (2010), since the structural parameters are set-identified under sign restrictions, they bring about robust conclusions across the set of SVARs satisfying the restrictions. However, sign restrictions often lead to overly agnostic results. In other words, they can result in very different implications for the IRFs from those predicted by economic theory. For example, Kilian and Murphy (2012) show that the values of the price elasticity of oil supply to demand shocks obtained from global oil market SVARs with only sign restrictions on the IRFs are disputable. Arias, Caldara, and Rubio-Ramírez (2019) also show

¹ Throughout the paper, ‘conditional’ refers to conditioning on all structural shocks other than the shock of interest being zero. It does not mean conditioning on the information available from past observations.

² It would be interesting to apply our correlation restriction to SVARs with other restrictions, such as recursive and long-run restrictions. However, that is beyond the scope of our paper, so we leave it for future research.

that, in the SVAR with sign restrictions on the IRFs of Uhlig (2005), the posterior estimates of the contemporaneous coefficients on the monetary policy equation are likely to violate the sign implied in the monetary rules embedded in dynamic stochastic general equilibrium (DSGE) models. Therefore, for sign-restricted SVARs, it is important to impose additional uncontroversial restrictions to reduce the set of the structural parameters that satisfy the sign restrictions. This is necessary in order to reach economically meaningful conclusions.

Our proposed correlation restriction helps identify structural shocks more precisely in SVARs with sign restrictions. Specifically, by adding an additional restriction on the conditional correlation between output and prices for demand or supply shocks based on economic theory, the correlation restriction narrows the set of the structural parameters that satisfy the sign restrictions, which enables the SVARs to lead to meaningful economic conclusions.

One difficulty in using correlations to identify demand and supply shocks is that output and price levels may be nonstationary. To handle this nonstationarity, researchers often estimate a stationary VAR using output growth and inflation rather than levels. The unconditional correlation between these stationary variables is well defined, but the corresponding levels need not have a time-invariant unconditional correlation. The same problem can arise when output and prices enter the VAR directly in levels and are nonstationary. Thus, relying on the usual unconditional correlation limits the restriction to variables for which it is well defined, such as stationary output growth and inflation.

However, the economic distinction between demand and supply shocks concerns whether output and price levels move in the same or opposite directions. Growth and inflation instead measure changes in these levels and can comove differently when output and prices adjust at different speeds. For example, after a contractionary demand shock, output may begin to recover while prices continue to fall. Growth and inflation then respond with opposite signs, even though both levels remain below their paths without the shock. After an adverse supply shock, output may likewise begin to recover while prices continue to rise. Growth and inflation then respond with the same sign, even though output remains below and prices above their paths without the shock. These differences in adjustment can make the correlation between growth and inflation differ in sign from that between output and price levels. We therefore need a measure of comovement between output and price levels that remains well defined even when the levels are nonstationary.

To this end, we build on Ernst et al. (2025), who analyze empirical correlations between nonstationary processes over finite intervals. We adapt their approach to the SVAR setting and use the expected sample correlation between output and price level paths over a finite period. Specifically, we construct paths implied by the SVAR for successive realizations of the shock of interest, with all other shocks set to zero. We calculate the sample correlation between each pair of level paths. We then average these correlations over the distribution of

shock sequences. Online Appendix A derives an analytical representation of this expectation, which we evaluate by numerical integration without simulating the paths.

Next, we show how to impose the correlation restriction jointly with sign restrictions on the IRFs and the structural parameters. Both types of restrictions can be expressed as inequalities in the structural parameters. Therefore, we can use the Bayesian methods developed in [Rubio-Ramírez, Waggoner, and Zha \(2010\)](#) and [Arias, Rubio-Ramírez, and Waggoner \(2018\)](#) for the correlation restriction without further modification. We also explain the Bayesian inference and the algorithm for sampling the structural parameters under both the sign restrictions and the correlation restriction.

Moreover, one might think that imposing sign restrictions on the IRFs of output and prices simultaneously would lead to the same results as the correlation restriction. Therefore, we compare the correlation restriction with the sign restrictions. We demonstrate that this is not the case, and that these two restrictions differ significantly. The key point is that the sign restrictions are imposed at each horizon, whereas the correlation restriction is imposed on the overall conditional correlation. For example, it may seem that sign restrictions requiring a positive demand shock to increase both output and prices over a sufficiently long period would ensure persistent positive comovement. However, even these extended sign restrictions need not imply a positive conditional correlation, as illustrated by the example in [Section 3.3](#). These sign restrictions also require pre-specifying the horizons at which output and prices must increase together. In contrast, the correlation restriction does not require pre-specifying the horizons at which the two variables increase (or decrease) together. Under the correlation restriction, the output and price IRFs can have opposite signs at some horizons, including on impact, as long as the overall conditional correlation is positive. Furthermore, under the sign restrictions, the signs of the IRFs of output and prices are restricted (for example, their signs must be positive for an expansionary demand shock). However, under the correlation restriction, the signs of the two IRFs are unrestricted. Their signs can either be positive or negative (even their signs can differ temporarily), as long as the overall conditional correlation between output and prices is positive. The advantage is to restrict their overall comovement directly without imposing additional signs on the individual responses.

Since monetary policy shocks are well-known demand shocks, we apply a correlation restriction for demand shocks to an SVAR for monetary policy shocks. As is well understood from economic theory, a contractionary monetary policy shock raises the nominal and real interest rates, which depresses aggregate demand. Accordingly, output and prices go down. Motivated by this comovement, we impose a correlation restriction that the conditional correlation between output and prices for monetary policy shocks must be positive in an SVAR model for monetary policy shocks. For the SVAR model, we build on that of [Arias, Caldara, and Rubio-Ramírez \(2019\)](#), in which sign restrictions on the structural parameters are im-

posed and the posterior distribution of price responses does not rule out the price puzzle.³ We show that the correlation restriction improves the identification of monetary policy shocks. More importantly, it makes the decline in prices clearer, helping to resolve the price puzzle.⁴

A good example of supply shocks is oil supply shocks, as the name of the shocks implies. Accordingly, we use a correlation restriction to identify oil supply shocks. As is well understood, an oil supply shock, which decreases world oil production, leads to an increase in the real oil price. Since oil is used in the production of almost all goods, the increased real oil price results in a rise in production costs, which depresses aggregate supply. Thus, output declines, but prices go up. This comovement motivates a negative conditional correlation restriction for oil supply shocks. For this application, an SVAR model for oil supply shocks based on [Baumeister and Peersman \(2013\)](#), in which sign restrictions on the IRFs are used, is considered.⁵ Specifically, we add a correlation restriction—the conditional correlation between output and prices for oil supply shocks must be negative—to the SVAR model for oil supply shocks. Estimation results clearly show that the correlation restriction helps oil supply shocks be sharply identified. With sign restrictions only, the 68 percent credible set for output includes zero at every displayed horizon. Adding the correlation restriction provides clearer evidence of a persistent decline in output and a more sustained increase in prices, consistent with economic theory.

One related issue raised by [Baumeister and Hamilton \(2024\)](#) is that, in set-identified SVARs, conventional Bayesian posterior summaries may be inappropriate. In particular, a given reduced form can generate multiple admissible structural paths through different rotations, and these paths have the same likelihood. The conventional practice based on a uniform prior over the rotation matrix can place some of these paths inside a reported credible set and others outside it, even though the likelihood does not distinguish among them. One can therefore either justify the prior used to rank these paths or report an object that does not condition on a particular prior over rotations. We follow [Giacomini and Kitagawa \(2021\)](#), which provides the latter approach, to examine the robustness of the identifying content of the correlation restriction. Specifically, we report the set of posterior means for the IRFs. This

³ In fact, [Arias, Caldara, and Rubio-Ramírez \(2019\)](#) impose zero restrictions, as well as the sign restrictions. For simplicity, we consider their SVAR with only the sign restrictions. However, in Online Appendix C, we also consider their SVAR with both the sign and zero restrictions, as in their identification scheme. Note that even when both the sign and zero restrictions are imposed in their SVAR, the price puzzle cannot be ruled out.

⁴ Some might think that the correlation restriction eliminates the price puzzle by definition. This is not the case. The correlation restriction does not rule out the structural parameters that lead to an increase in output and prices in response to a contractionary monetary policy shock. Furthermore, as previously mentioned, under the correlation restriction, output and prices can react in opposite directions, even during the first few periods after the shock.

⁵ Different from [Baumeister and Peersman \(2013\)](#), we do not allow the parameters to be time-varying because the focus of this paper is not the time-varying effects of oil supply shocks.

set gives the lower and upper bounds on posterior mean responses attainable by rotations that satisfy the identifying restrictions. We then compare the set under the sign restrictions and under the sign and correlation restrictions to assess whether the correlation restriction adds identifying content beyond the sign restrictions. The results show that the correlation restriction rules out a sizable part of the rotations admitted by the sign restrictions and narrows the set of posterior means by removing responses that are less consistent with the economic interpretations of monetary policy and oil supply shocks.

This paper is related to three strands of literature. The first one is studies on SVARs with sign restrictions on the IRFs or the structural parameters. Since [Faust \(1998\)](#), [Canova and De Nicoló \(2002\)](#), and [Uhlig \(2005\)](#), sign restrictions on the IRFs or the structural parameters in SVARs have been frequently used in many contexts of empirical studies, including not only monetary policy shocks and oil shocks, as in our study, but also government spending shocks, technology shocks, and financial shocks.⁶

The second strand is studies on the effects of monetary policy shocks using SVARs. For instance, [Christiano, Eichenbaum, and Evans \(1996\)](#) use classic zero restrictions (i.e., the recursive restrictions). Since their seminal work, various methods to identify monetary policy shocks, such as factor-augmented VARs ([Bernanke, Boivin, and Elias, 2005](#)), sign restrictions ([Faust, 1998](#); [Canova and De Nicoló, 2002](#); [Uhlig, 2005](#); and [Arias, Caldara, and Rubio-Ramírez, 2019](#)), and high-frequency identification and proxy SVARs ([Gertler and Karadi, 2015](#); and [Jarociński and Karadi, 2020](#)), etc., have been used in SVARs.

The last strand is studies on the effects of oil shocks using SVARs. [Kilian \(2009\)](#) estimates an SVAR with zero restrictions to disentangle oil supply, aggregate demand, and oil-specific demand shocks. Since his seminal work, researchers have used a variety of approaches to identify oil shocks within the SVAR framework, which includes sign restrictions ([Kilian and Murphy, 2012](#); [Lippi and Nobili, 2012](#); [Peersman and Van Robays, 2012](#); [Baumeister and Peersman, 2013](#); [Baumeister and Hamilton, 2019](#); and [Kim, 2020](#)), institutional features of OPEC and high-frequency data ([Känzig, 2021](#)), and narrative information ([Antolín-Díaz and Rubio-Ramírez, 2018](#); and [Caldara, Cavallo, and Iacoviello, 2019](#)).

Among the studies reviewed above, [Canova and De Nicoló \(2002\)](#) is particularly close to our paper because both papers identify structural shocks through the comovement they generate. They impose sign restrictions on conditional cross-correlations in a stationary VAR. We instead restrict the expected sample correlation between shock-generated paths over

⁶ For example, [Faust \(1998\)](#), [Canova and De Nicoló \(2002\)](#), [Uhlig \(2005\)](#), [Amir-Ahmadi and Uhlig \(2015\)](#), and [Arias, Caldara, and Rubio-Ramírez \(2019\)](#) for monetary policy shocks, [Pappa \(2009\)](#) and [Mountford and Uhlig \(2009\)](#) for government spending shocks, [Dedola and Neri \(2007\)](#) and [Peersman and Straub \(2009\)](#) for technology shocks, [Kilian and Murphy \(2012\)](#), [Baumeister and Peersman \(2013\)](#), [Baumeister and Hamilton \(2019\)](#), and [Kim \(2020\)](#) for oil shocks, and [Hristov, Hülsewig, and Wollmershäuser \(2012\)](#) and [Gambetti and Musso \(2017\)](#) for financial shocks.

a specified finite period. We derive an analytical representation of this expectation from the variables' IRFs without requiring unconditional covariances to exist. The restriction can therefore be applied to both stationary and nonstationary variables, including log output and log prices whose unconditional covariances may not exist. It also allows researchers to target comovement over a period of economic interest, even when unconditional correlations are well defined.

A further difference concerns inference over admissible rotations. [Canova and De Nicoló \(2002\)](#) search over a finite grid and select an orthogonal decomposition using the signs of conditional cross-correlations. In contrast, we impose the correlation restriction by truncating the prior support of the structural parameters. We then integrate over admissible rotations in Bayesian inference rather than selecting a single decomposition, thereby accounting for uncertainty over the rotation matrix.

Taken together, the above literature has developed a variety of identifying strategies for SVARs. This paper contributes to this literature by introducing the *correlation restriction* into Bayesian SVAR inference as an additional economically plausible source of structural identification. Motivated by the economic distinction between demand and supply shocks, the restriction requires that the conditional correlation between output and prices for demand (supply) shocks be positive (negative). Importantly, the correlation restriction does not replace existing identifying restrictions. Rather, it can be imposed jointly with sign, zero, or narrative sign restrictions and provides an additional source of identifying information through a theory-consistent restriction on conditional comovement. We show that this additional restriction is useful in identifying monetary policy shocks and oil supply shocks by narrowing the admissible set of structural parameters and sharpening posterior inference.

The paper proceeds as follows. Section 2 presents the basic SVAR framework for the correlation restriction as well as sign restrictions on the IRFs or the structural parameters, and defines the conditional correlation between output and prices for demand or supply shocks. Section 3 expresses the correlation restriction and sign restrictions as inequalities in the structural parameters and compares the two types of restrictions. Section 4 explains the Bayesian inference for the SVAR with the correlation and sign restrictions, and describes the algorithm for sampling the structural parameters under the sign restrictions and the correlation restriction for demand shocks. Section 5 applies the correlation restriction to identify monetary policy shocks and oil supply shocks and reports the robust Bayesian inference results. Section 6 concludes the paper.

2 SVAR framework

Consider the general form of an SVAR:

$$\mathbf{Y}'_t \mathbf{A} = \sum_{j=1}^p \mathbf{Y}'_{t-j} \mathbf{C}_j + \boldsymbol{\alpha} + \boldsymbol{\varepsilon}'_t, \quad (1)$$

where $\mathbf{Y}_t$ is an $n \times 1$ vector of endogenous variables, $\mathbf{A}$ and $\mathbf{C}_j$ ($j = 1, \dots, p$) are $n \times n$ parameter matrices, $\boldsymbol{\alpha}$ is a $1 \times n$ vector of constants, p is the lag length, T is the sample size, and $\boldsymbol{\varepsilon}_t$ is an $n \times 1$ vector of serially independent Gaussian structural shocks with mean zero and covariance matrix $\mathbf{I}_n$ (the $n \times n$ identity matrix). Equation (1) can be written more compactly as

$$\mathbf{Y}'_t \mathbf{A} = \mathbf{X}'_t \boldsymbol{\Gamma} + \boldsymbol{\varepsilon}'_t, \quad (2)$$

where $\boldsymbol{\Gamma}' = [\mathbf{C}'_1, \dots, \mathbf{C}'_p, \boldsymbol{\alpha}']$ is an $n \times m$ ($m = np + 1$) matrix of parameters, and $\mathbf{X}'_t = [\mathbf{Y}'_{t-1}, \dots, \mathbf{Y}'_{t-p}, 1]$ is a $1 \times m$ vector consisting of lagged endogenous variables and the constant term. Multiplying both sides of Equation (2) by $\mathbf{A}^{-1}$ gives the reduced-form representation:

$$\mathbf{Y}'_t = \mathbf{X}'_t \mathbf{B} + \mathbf{u}'_t, \quad (3)$$

where $\mathbf{B} = \boldsymbol{\Gamma} \mathbf{A}^{-1}$ and $\mathbf{u}'_t = \boldsymbol{\varepsilon}'_t \mathbf{A}^{-1}$. Here, $\mathbf{u}_t$ denotes the vector of reduced-form shocks, and its covariance matrix is $\mathbb{E}_t[\mathbf{u}_t \mathbf{u}'_t] = (\mathbf{A} \mathbf{A}')^{-1} = \boldsymbol{\Sigma}$.

2.1 Impulse response functions

Let the response of the i th variable to the k th structural shock at horizon h be the element in row i and column k of $\mathbf{IR}_h(\boldsymbol{\Psi})$, where $\boldsymbol{\Psi} = (\mathbf{A}, \boldsymbol{\Gamma})$ collects the structural parameters. Then, $\mathbf{IR}_h(\boldsymbol{\Psi})$ is defined by

$$\mathbf{IR}_0(\boldsymbol{\Psi}) = (\mathbf{A}^{-1})' \quad (4)$$

$$\text{and } \mathbf{IR}_h(\boldsymbol{\Psi}) = \sum_{j=1}^{\min(h,p)} (\mathbf{C}_j \mathbf{A}^{-1})' \mathbf{IR}_{h-j}(\boldsymbol{\Psi}), \text{ for } h > 0. \quad (5)$$

Equations (4) and (5) clearly show that the IRFs depend only on the structural parameters. Moreover, from Equation (4), the contemporaneous response, $\mathbf{IR}_0$, of the i th variable to the k th structural shock is the (i, k) th element of $(\mathbf{A}^{-1})'$, one of the structural parameters. Hence, restricting the element in the (i, k) th element of $(\mathbf{A}^{-1})'$ to be positive (negative) is identical to restricting the initial IRF of the i th variable to the k th structural shock to be positive (negative).

2.2 Conditional correlations between output and prices

Let y_t and p_t denote the log levels of output and prices, and let ε_t^k denote a demand shock ($k = d$) or a supply shock ($k = s$). We write their log-level responses to shock k as $\mathbf{IR}_h^{(y,k)}(\Psi)$ and $\mathbf{IR}_h^{(p,k)}(\Psi)$. When output and prices enter the VAR in log differences, these responses are obtained by cumulating the corresponding entries of $\mathbf{IR}_h(\Psi)$ in Equations (4) and (5).

Suppose we measure the comovement of output and prices over $H_c \geq 2$ periods, where H_c denotes the correlation horizon. To do so, consider the log-level paths $\tilde{y}_{t+r}^k$ and $\tilde{p}_{t+r}^k$, $r = 1, \dots, H_c$, generated by successive shocks of type k . Each path is measured relative to the corresponding path with no shocks after date t and the same initial history. All other structural shocks are set to zero. Shocks of type k remain random. Let $\bar{y}^k = H_c^{-1} \sum_{r=1}^{H_c} \tilde{y}_{t+r}^k$ and $\bar{p}^k = H_c^{-1} \sum_{r=1}^{H_c} \tilde{p}_{t+r}^k$ be their respective means over this period. We define the conditional correlation as the expected sample correlation between these paths:

$$\rho_{y,p}^k(\Psi; H_c) = E_{\varepsilon^k} \left[\frac{\sum_{r=1}^{H_c} (\tilde{y}_{t+r}^k - \bar{y}^k)(\tilde{p}_{t+r}^k - \bar{p}^k)}{\sqrt{\sum_{r=1}^{H_c} (\tilde{y}_{t+r}^k - \bar{y}^k)^2} \sqrt{\sum_{r=1}^{H_c} (\tilde{p}_{t+r}^k - \bar{p}^k)^2}} \right], \quad k \in \{d, s\}. \quad (6)$$

Here, ‘conditional’ refers to conditioning on all other structural shocks being zero. It does not refer to conditioning on observations available at date t . The expectation is over Gaussian shock sequences, holding Ψ fixed. Thus, Equation (6) measures comovement throughout the period.

A simple plug-in approximation to Equation (6) replaces the sample moments by their model-implied expectations.⁷ In particular, the sample means $\bar{y}^k$ and $\bar{p}^k$ are replaced by zero, and the averages of squares and cross products are replaced by their expectations. This gives

$$\rho_{y,p}^k(\Psi; H_c) \approx \frac{\sum_{h=0}^{H_c-1} (H_c - h) \mathbf{IR}_h^{(y,k)}(\Psi) \mathbf{IR}_h^{(p,k)}(\Psi)}{\sqrt{\sum_{h=0}^{H_c-1} (H_c - h) \left(\mathbf{IR}_h^{(y,k)}(\Psi) \right)^2} \sqrt{\sum_{h=0}^{H_c-1} (H_c - h) \left(\mathbf{IR}_h^{(p,k)}(\Psi) \right)^2}}. \quad (7)$$

However, substituting expected moments into the correlation formula need not give the expected sample correlation. The approximation can even have the opposite sign.⁸ To impose

⁷ For comparison, with Ψ fixed, the correlation between $\tilde{y}_{t+H_c}^k$ and $\tilde{p}_{t+H_c}^k$ at the single date $t + H_c$ is

$$\frac{\sum_{h=0}^{H_c-1} \mathbf{IR}_h^{(y,k)}(\Psi) \mathbf{IR}_h^{(p,k)}(\Psi)}{\sqrt{\sum_{h=0}^{H_c-1} \left(\mathbf{IR}_h^{(y,k)}(\Psi) \right)^2} \sqrt{\sum_{h=0}^{H_c-1} \left(\mathbf{IR}_h^{(p,k)}(\Psi) \right)^2}}.$$

This expression measures the correlation between output and prices at $t + H_c$ only. In contrast, Equation (6) computes their sample correlation using all H_c observations from $t + 1$ through $t + H_c$ and then takes its expectation.

⁸ For example, let $H_c = 2$ and suppose that cumulating the IRFs from a stable VAR in log differences

the restriction on Equation (6), we therefore evaluate the expectation itself rather than this approximation.

Following Ernst et al. (2025), who study the moments of empirical correlations between Brownian motion paths over finite intervals, we derive in Online Appendix A the exact analytical representation

$$\rho_{y,p}^k(\Psi; H_c) = \frac{2}{\pi} \int_0^\infty \int_0^\infty \frac{\text{tr}[\mathbf{K}_{yp}^k \mathbf{D}^k(u, v)^{-1}]}{\sqrt{\det \mathbf{D}^k(u, v)}} du dv, \quad (8)$$

where $\mathbf{D}^k(u, v) = \mathbf{I}_{H_c} + u^2 \mathbf{K}_{yy}^k + v^2 \mathbf{K}_{pp}^k$. The matrices $\mathbf{K}_{yy}^k$, $\mathbf{K}_{pp}^k$, and $\mathbf{K}_{yp}^k$ are constructed as follows. For each variable, we first arrange its level IRFs in an $H_c \times H_c$ lower triangular matrix, with the impact response on the main diagonal and later responses on successive lower diagonals. We then subtract the mean of each column. Multiplying each centered matrix on the left by its transpose gives $\mathbf{K}_{yy}^k$ and $\mathbf{K}_{pp}^k$ for output and prices, respectively. To obtain $\mathbf{K}_{yp}^k$, we multiply the transpose of the centered output matrix by the centered price matrix and average this product with its transpose. Online Appendix A provides the details.

We use numerical integration of Equation (8) to impose the correlation restriction. This does not require simulating shock sequences. The integral is well defined whenever both sample variances are positive with probability one, even if output and prices are nonstationary in levels. For a fixed H_c , it depends only on Ψ ; we suppress H_c in the notation below.

Allowing for nonstationary levels is important here because our correlation restriction concerns the comovement of output and price levels, which distinguishes demand from supply shocks. In our applications, however, output and prices enter the VAR in log differences, and we restrict the reduced-form prior to stable VARs. Even under this restriction, the implied log levels can be nonstationary, so their unconditional covariances may not exist. Equation (6) avoids the need for these covariances by measuring comovement within a specified finite period.

Choice of the correlation horizon The choice of H_c determines how much of the adjustment process this measure captures. A short horizon may mainly reflect initial responses and miss delayed adjustments, especially when output and prices respond at different speeds. Extending the horizon captures more of these adjustments, but a longer horizon is not necessarily preferable. When shocks have permanent effects on levels, their accumulated effects can dominate variation over a long observation window. The correlation may then reflect these lasting effects more than comovement during adjustment. A finite horizon limits this influence without removing permanent effects from the responses.

gives log-level responses of (1,0.2) for output and (1,3) for prices at horizons zero and one. The plug-in approximation is 0.549, whereas our expected sample correlation is -0.134 .

These considerations call for choosing H_c according to the period of economic interest, rather than increasing it solely until the estimated results cease to change. To capture initial responses and delayed adjustments in output and prices while limiting the influence of accumulated permanent effects, we use a five-year correlation horizon as a practical compromise in both applications. Different horizons may be more appropriate in other applications. Online Appendix F examines sensitivity to this choice using three- and seven-year horizons.

If the objective is instead to approximate the unconditional correlation between stationary components driven by a given shock, H_c can be chosen sufficiently large. Under stable dynamics and positive stationary variances, initialization effects vanish as the sample grows, and sample moments converge to their population counterparts. The expected sample correlation in Equation (6) therefore converges to the corresponding unconditional correlation as $H_c \rightarrow \infty$. The horizon needed for a close approximation depends on the persistence of the two variables. This convergence result does not apply to nonstationary level components.

3 Identification via sign and correlation restrictions

The structural parameters in Equation (1) are not identified, so additional restrictions are required for identification. This paper proposes a new approach that uses the conditional correlation between output and prices for demand or supply shocks: the correlation restriction. Our approach builds on the existing SVAR literature that imposes sign restrictions on the IRFs and the structural parameters, including [Canova and De Nicoló \(2002\)](#), [Uhlig \(2005\)](#), [Baumeister and Peersman \(2013\)](#), and [Arias, Caldara, and Rubio-Ramírez \(2019\)](#), among others.

The correlation restriction imposes economically plausible comovement between output and prices in the SVAR by requiring the structural parameters to deliver the appropriate sign of the conditional correlation between the two variables for each type of shock. In particular, demand shocks are associated with positive comovement, so output and prices driven by demand shocks tend to move in the same direction. A contractionary monetary policy shock, for example, raises nominal and real interest rates, reduces consumption and investment, and depresses aggregate demand. Consequently, both output and prices fall.

Supply shocks, by contrast, are associated with negative comovement between output and prices. For instance, an oil supply shock (reducing world oil production) raises oil prices. Oil prices are included directly in the consumer price index (CPI), and higher oil prices also increase production costs. As a result, prices rise while output falls. Given these economic intuitions, researchers wish to restrict the structural parameters so that the correlation between output and prices is positive when only demand shocks occur and negative when only supply shocks occur.

3.1 Sign restrictions

Rubio-Ramírez, Waggoner, and Zha (2010) and Arias, Rubio-Ramírez, and Waggoner (2018) show that sign restrictions can be written as

$$\Upsilon(\Psi) = (\mathbf{e}'_{1,n} \mathbf{F}(\Psi)' \mathbf{S}'_1, \dots, \mathbf{e}'_{n,n} \mathbf{F}(\Psi)' \mathbf{S}'_n)' > \mathbf{0}, \quad (9)$$

where $\mathbf{e}_{k,n}$ is the k th column of the $n \times n$ identity matrix.

The definitions of $\mathbf{F}(\Psi)$ and $\mathbf{S}_k$ depend on whether the sign restrictions are imposed on the IRFs or on the structural parameters. When imposing sign restrictions on the IRFs, $\mathbf{F}(\Psi)$ stacks the IRFs vertically at the relevant horizons, and $\mathbf{S}_k$ selects the horizons and variables to which the sign restrictions are applied. When imposing sign restrictions on the structural parameters, $\mathbf{F}(\Psi) = \Psi$, and $\mathbf{S}_k$ selects the entries of Ψ on which the sign restrictions are imposed.

3.2 Correlation restriction

Sign restrictions are inequality restrictions and are therefore often viewed as less stringent than exact zero restrictions. A drawback, however, is that sign restrictions alone may leave a large set of admissible rotations, and hence wide credible sets for the IRFs. The correlation restriction provides an additional, economically interpretable restriction that targets the comovement between output and prices. As a result, it can shrink the set of admissible rotations and sharpen inference on the dynamic responses. Thus, we identify demand and supply shocks by combining sign restrictions on the IRFs or the structural parameters with the correlation restriction.

Specifically, conditional on all structural shocks other than demand shocks being zero, we expect a positive correlation between output and prices. We therefore impose

$$\rho_{y,p}^d(\Psi) > 0. \quad (10)$$

Conversely, conditional on all structural shocks other than supply shocks being zero, we expect a negative correlation between output and prices. We therefore impose

$$\rho_{y,p}^s(\Psi) < 0. \quad (11)$$

The restriction concerns the expected sample correlation of the shock-generated level paths over the specified period H_c , rather than an unconditional correlation of potentially nonstationary levels.

Discussion for the size of the conditional correlations Some might consider this restriction on the conditional correlation to be too weak because many researchers typically consider correlations within the range of -0.2 to 0.2 to be meaningless. Accordingly, they might wish to impose a stronger restriction on the conditional correlation than the baseline restrictions in Equations (10) and (11). In this case, we can impose $\rho_{y,p}^d(\Psi) > 0.2$ or $\rho_{y,p}^s(\Psi) < -0.2$. We refer to this as the strong correlation restriction.

The effect of the strong correlation restriction need not be uniform and may vary across datasets. In our case, to examine how imposing the strong correlation restriction affects the estimation results, Online Appendix B reports results under the strong correlation restriction. These estimation results are nearly identical to those under the baseline correlation restrictions.

3.3 Comparison of the correlation restriction with sign restrictions

One might think that requiring a positive demand shock to increase output and prices over a sufficiently long period could replace the correlation restriction. However, extending sign restrictions does not, in general, ensure the required positive conditional correlation. This is because the sign of the expected sample correlation depends on the relative magnitudes and timing of the responses, not just their signs.

For expositional clarity, we focus our discussion on the identification of demand shocks. Suppose a researcher seeks to obtain a positive conditional correlation between output and prices through sign restrictions. For intuition, consider the plug-in approximation in Equation (7), which is positive when its covariance term is positive:

$$\sum_{h=0}^{H_c-1} (H_c - h) \mathbf{IR}_h^{(y,d)}(\Psi) \mathbf{IR}_h^{(p,d)}(\Psi) > 0. \quad (12)$$

To satisfy Equation (12), suppose the researcher imposes two sign restrictions requiring a positive (expansionary) demand shock to increase both variables at prespecified horizons:

$$\begin{aligned} & \mathbf{IR}_h^{(y,d)}(\Psi) > 0 \\ \text{and } & \mathbf{IR}_h^{(p,d)}(\Psi) > 0, \text{ for } h = 0, \dots, H_c - 1, \end{aligned}$$

where, for comparison, the restrictions cover all horizons entering the H_c -period paths. These restrictions satisfy Equation (12) because every term in the sum is positive. Even if these restrictions yield a positive expected sample correlation, they still rule out initially zero or negative responses and sign reversals within the restricted period.

By contrast, the correlation restriction does not require each IRF to be positive. The

plug-in condition in Equation (12) illustrates this distinction: only the sum must be positive, not the product at every horizon. The IRFs can temporarily move in opposite directions while the sum remains positive.

Even when both IRFs are positive at every horizon entering the H_c -period paths, the expected sample correlation in Equation (6) can be negative. For a simple illustration, consider the log-level IRFs

$$\begin{aligned}\mathbf{IR}_h^{(y,d)}(\Psi) &= 0.2^h, \\ \mathbf{IR}_h^{(p,d)}(\Psi) &= 5 - 4(0.5)^h, \quad h = 0, 1, \dots\end{aligned}$$

Both level responses are strictly positive at every horizon. With $H_c = 20$, evaluating the integral in Equation (8) gives $\rho_{y,p}^d(\Psi) \approx -0.156$. Extending the correlation horizon to $H_c = 60$ still yields a negative expected sample correlation of -0.098 . Positive level IRFs mean that output and prices lie above their respective paths without the shock. The sample correlation instead measures comovement around each path's own time mean. Thus, the two restrictions constrain different features of the responses.

The distinction between the two restrictions remains even when the sign restrictions yield a positive conditional correlation between output and prices for demand shocks. The correlation restriction in Equation (10) requires only a positive expected sample correlation between the output and price paths generated by successive demand shocks. It does not fix the signs of their IRFs. That is, the signs of the output and price IRFs can either be positive or negative (even their signs can differ temporarily), as long as the overall conditional correlation between output and prices is positive. By contrast, the sign restrictions require both output and prices to rise at prespecified horizons after a demand shock. Thus, although the correlation restriction requires a choice of H_c , it does not prescribe the individual horizons at which the two variables must move in the same direction. This is useful when theory gives a clearer prediction about overall comovement than about the timing and sign of each individual response.

The correlation restriction therefore allows researchers to restrict overall comovement directly while leaving the signs of the individual responses unrestricted. It imposes positive comovement between output and prices as an additional identifying assumption motivated by the demand shock interpretation. For example, in the monetary policy application, we combine sign restrictions on the policy equation with the correlation restriction, without requiring output or prices to fall after a contractionary shock. This allows us to examine the price puzzle without imposing a negative price response.

4 Bayesian inference

This section shows how to impose the correlation restriction in the SVAR in which sign restrictions are already imposed. Our approach extends the Bayesian methodology of [Rubio-Ramírez, Waggoner, and Zha \(2010\)](#), [Arias, Rubio-Ramírez, and Waggoner \(2018\)](#), and [Antolín-Díaz and Rubio-Ramírez \(2018\)](#). For brevity, we focus on the correlation restriction for demand shocks, $\rho_{y,p}^d(\Psi) > 0$. The correlation restriction for supply shocks, $\rho_{y,p}^s(\Psi) < 0$, can be handled analogously.

4.1 Prior distribution

We begin by specifying a prior for the reduced-form parameters $(\mathbf{B}, \Sigma)$. We assign a conjugate Normal-Inverse-Wishart prior, $\mathcal{NIW}(\nu, \Phi, \mathbf{S}, \Omega)$. Specifically,

$$\begin{aligned} \text{vec}(\mathbf{B}) \mid \Sigma &\sim \mathcal{N}(\text{vec}(\mathbf{S}), \Sigma \otimes \Omega) \\ \text{and } \Sigma &\sim \mathcal{IW}(\nu, \Phi). \end{aligned}$$

Our study uses an improper, noninformative prior with ν , Φ , $\mathbf{S}$, and Ω^{-1} set to zero (or zero matrices, as appropriate), following [Arias, Caldara, and Rubio-Ramírez \(2019\)](#), to ensure that the estimation results are driven primarily by the data rather than the prior.

In a Bayesian SVAR, sign restrictions are implemented by truncating the prior distribution for the reduced-form parameters and the rotation matrix. The correlation restriction is implemented in the same way. However, as Equations (9) and (10) show, these two restrictions are stated in terms of the structural parameters, rather than the reduced-form parameters. Therefore, to truncate this prior using these two restrictions, we first need to derive a mapping between the structural parameters and the reduced-form parameters together with the rotation matrix, and to rewrite the restrictions as functions of the reduced-form parameters and the rotation matrix. This makes it clear which region of the prior distribution satisfies the restrictions.

To derive the mapping, we introduce the set $\mathcal{O}(n)$ of $n \times n$ orthogonal matrices and an orthogonal rotation matrix $\mathbf{Q} \in \mathcal{O}(n)$ that links the structural shocks to the reduced-form shocks. Expressing the reduced-form shocks in Equation (3) as rotations of the structural shocks via $\mathbf{Q}$ yields

$$\mathbf{Y}'_t = \mathbf{X}'_t \mathbf{B} + \varepsilon'_t \mathbf{Q}' R(\Sigma), \quad (13)$$

where $R(\Sigma)$ is an $n \times n$ matrix satisfying $R(\Sigma)' R(\Sigma) = \Sigma$. A Cholesky decomposition is typically used for $R(\cdot)$.

Building on Equations (2) and (13), [Arias, Rubio-Ramírez, and Waggoner \(2018\)](#) and

Antolín-Díaz and Rubio-Ramírez (2018) show that the mapping between the structural parameters and the reduced-form parameters together with the rotation matrix is given by

$$f_R(\Psi) = \left(\underbrace{\Gamma \mathbf{A}^{-1}}_{\mathbf{B}}, \underbrace{(\mathbf{A} \mathbf{A}')^{-1}}_{\mathbf{\Sigma}}, \underbrace{R((\mathbf{A} \mathbf{A}')^{-1}) \mathbf{A}}_{\mathbf{Q}} \right) \quad (14)$$

and that its inverse is given by

$$f_R^{-1}(\mathbf{B}, \mathbf{\Sigma}, \mathbf{Q}) = \left(\underbrace{R(\mathbf{\Sigma})^{-1} \mathbf{Q}}_{\mathbf{A}}, \underbrace{\mathbf{B} R(\mathbf{\Sigma})^{-1} \mathbf{Q}}_{\mathbf{\Gamma}} \right). \quad (15)$$

Equation (15) implies that the two restrictions can be expressed in terms of the reduced-form parameters and $\mathbf{Q}$.

Specifically, the sign restrictions in Equation (9) can be written as

$$\Upsilon(f_R^{-1}(\mathbf{B}, \mathbf{\Sigma}, \mathbf{Q})) > 0.$$

Under a uniform prior on $\mathbf{Q}$ over $\mathcal{O}(n)$,⁹ the sign restrictions truncate the prior distribution to the subset of the parameter space that satisfies the above inequality. The resulting prior distribution is given by

$$\pi(\mathbf{B}, \mathbf{\Sigma}, \mathbf{Q} \mid \Upsilon(f_R^{-1}(\mathbf{B}, \mathbf{\Sigma}, \mathbf{Q})) > 0) \propto \mathcal{N}\mathcal{I}\mathcal{W}(\nu, \mathbf{\Phi}, \mathbf{S}, \mathbf{\Omega}) \mathbb{I}[\Upsilon(f_R^{-1}(\mathbf{B}, \mathbf{\Sigma}, \mathbf{Q})) > 0],$$

where $\mathbb{I}[\cdot]$ denotes an indicator function that equals 1 if the argument is true and 0 otherwise.

Similarly, the correlation restriction for demand shocks, Equation (10), can be expressed as

$$\rho_{y,p}^d(f_R^{-1}(\mathbf{B}, \mathbf{\Sigma}, \mathbf{Q})) > 0,$$

and the associated truncated prior is given by

$$\pi(\mathbf{B}, \mathbf{\Sigma}, \mathbf{Q} \mid \rho_{y,p}^d(f_R^{-1}(\mathbf{B}, \mathbf{\Sigma}, \mathbf{Q})) > 0) \propto \mathcal{N}\mathcal{I}\mathcal{W}(\nu, \mathbf{\Phi}, \mathbf{S}, \mathbf{\Omega}) \mathbb{I}[\rho_{y,p}^d(f_R^{-1}(\mathbf{B}, \mathbf{\Sigma}, \mathbf{Q})) > 0].$$

Moreover, when the correlation restriction is imposed jointly with the sign restrictions, the prior is truncated to

$$\mathcal{N}\mathcal{I}\mathcal{W}(\nu, \mathbf{\Phi}, \mathbf{S}, \mathbf{\Omega}) \mathbb{I}[\Upsilon(f_R^{-1}(\mathbf{B}, \mathbf{\Sigma}, \mathbf{Q})) > 0] \mathbb{I}[\rho_{y,p}^d(f_R^{-1}(\mathbf{B}, \mathbf{\Sigma}, \mathbf{Q})) > 0].$$

⁹ Baumeister and Hamilton (2015) show that a uniform prior over $\mathbf{Q}$ can induce nonuniform marginal priors for individual IRFs. For joint inference, however, Arias, Rubio-Ramírez, and Waggoner (2025) show that, conditional on the reduced-form parameters, this prior induces a uniform joint prior over the identified set for the vector of IRFs.

In our applications, we further restrict the prior support to stable VARs by requiring all eigenvalues of the reduced-form companion matrix to have modulus less than one. This condition applies to the variables as entered in the VAR, not to the implied log levels used to compute the correlation.

4.2 Posterior distribution

We first derive the posterior distribution in the absence of sign and correlation restrictions. Since $\mathcal{N}\mathcal{I}\mathcal{W}$ is conjugate to the likelihood function of the SVAR, the posterior distribution also has a closed form $\mathcal{N}\mathcal{I}\mathcal{W}$ representation (Arias, Rubio-Ramírez, and Waggoner, 2018). In particular, the posterior is

$$\begin{aligned} & \mathcal{N}\mathcal{I}\mathcal{W}(\hat{\nu}, \hat{\Phi}, \hat{\mathbf{S}}, \hat{\Omega}), \\ & \text{with } \hat{\nu} = T + \nu, \\ & \hat{\Omega} = (\mathbf{X}'\mathbf{X} + \Omega^{-1})^{-1}, \\ & \hat{\mathbf{S}} = \hat{\Omega}(\mathbf{X}'\mathbf{Y} + \Omega^{-1}\mathbf{S}), \\ & \text{and } \hat{\Phi} = \mathbf{Y}'\mathbf{Y} + \Phi + \mathbf{S}'\Omega^{-1}\mathbf{S} - \hat{\mathbf{S}}'\hat{\Omega}^{-1}\hat{\mathbf{S}}, \end{aligned} \tag{16}$$

for $\mathbf{Y} = [\mathbf{Y}_1 \cdots \mathbf{Y}_T]'$ and $\mathbf{X} = [\mathbf{X}_1 \cdots \mathbf{X}_T]'$.

When restrictions are imposed via a truncated prior, posterior simulation remains straightforward, provided that the posterior distribution without the restrictions is available in closed form. In particular, we use an acceptance-rejection sampler that draws candidate values from this closed form posterior and retains only those that satisfy the sign and correlation restrictions. The following algorithm provides a concrete description of how our sampling procedure for the structural parameters operates when both the sign restrictions and the correlation restriction are imposed.

Algorithm 1: Sampling the structural parameters under the sign restrictions and the correlation restriction for demand shocks

- (i) Draw $(\mathbf{B}, \Sigma)$ independently from the $\mathcal{N}\mathcal{I}\mathcal{W}$ posterior distribution in Equation (16).
- (ii) Draw $\mathbf{Q}$ independently from the uniform distribution over $\mathcal{O}(n)$.
- (iii) Based on the sampled $(\mathbf{B}, \Sigma, \mathbf{Q})$, check whether the sign restrictions in Equation (9) are satisfied. If the sign restrictions are satisfied (i.e., $\mathbb{I}[\Upsilon(f_R^{-1}(\mathbf{B}, \Sigma, \mathbf{Q})) > 0] = 1$), proceed to Step (iv) and otherwise return to Step (i).
- (iv) Compute the log-level IRFs of output and prices to demand shocks for $h = 0, \dots, H_c - 1$ and evaluate Equation (8) to obtain $\rho_{y,p}^d(\Psi)$. If the correlation restriction is satisfied

(i.e., $\mathbb{I}[\rho_{y,p}^d(f_R^{-1}(\mathbf{B}, \mathbf{\Sigma}, \mathbf{Q})) > 0] = 1$), proceed to Step (v) and otherwise return to Step (i).

- (v) Using Equation (15), transform $(\mathbf{B}, \mathbf{\Sigma}, \mathbf{Q})$ into $\mathbf{\Psi}$. If the required number of draws of the structural parameters $\mathbf{\Psi}$ has been achieved, terminate the algorithm and otherwise return to Step (i).

5 Applications

In this section, we use the correlation restriction to identify monetary policy and oil supply shocks. Monetary policy shocks are a type of demand shock, and oil supply shocks are, as the name suggests, a type of supply shock. Therefore, in the former case, we impose a correlation restriction that the conditional correlation between output and prices for monetary policy shocks must be positive. For identifying oil supply shocks, we impose a correlation restriction that the conditional correlation between output and prices for oil supply shocks must be negative. Using the two correlation restrictions, we first identify monetary policy shocks and then oil supply shocks.

5.1 Monetary policy shocks

The most typical example of demand shocks is monetary policy shocks. As is well-known, a contractionary monetary policy shock raises nominal and real interest rates, which depresses aggregate demand. Accordingly, output and prices fall in response to the shock. This comovement motivates a positive conditional correlation restriction. Hence, in this subsection, we use this feature of monetary policy shocks to identify monetary policy shocks more precisely.

Specifically, we use an SVAR based on [Arias, Caldara, and Rubio-Ramírez \(2019\)](#), which does a relatively good job in identifying monetary policy shocks. We add the correlation restriction for demand shocks to this SVAR. Then, we show that when adding the correlation restriction, a contractionary monetary policy shock decreases both output and prices (helping to resolve the price puzzle).

5.1.1 Data and others

The SVAR model for monetary policy shocks includes six monthly US variables, as in [Arias, Caldara, and Rubio-Ramírez \(2019\)](#): real GDP, the GDP deflator, a commodity price index, total reserves, nonborrowed reserves, and the federal funds rate. The monthly real GDP and GDP deflator, constructed by interpolation of the corresponding quarterly time series, are taken from [Arias, Caldara, and Rubio-Ramírez \(2019\)](#). The source of the commodity price

index is Global Financial Data, and total reserves, nonborrowed reserves, and the federal funds rate are obtained from the Federal Reserve Economic Data (FRED). All variables are seasonally adjusted (with the exception of the commodity price index and the federal funds rate). When estimating, we use the first log difference of all the variables (with the exception of the federal funds rate, which enters in percent), multiplied by 100, following [Arias, Caldara, and Rubio-Ramírez \(2019\)](#).¹⁰

Following [Arias, Caldara, and Rubio-Ramírez \(2019\)](#) and [Antolín-Díaz and Rubio-Ramírez \(2018\)](#), we end the sample in 2007 to limit the influence of the global financial crisis and unconventional monetary policy. Our sample runs from January 1965 to November 2007, excluding the large movements in reserves that began in December 2007. The SVAR includes 12 lags and a constant term, as in [Arias, Caldara, and Rubio-Ramírez \(2019\)](#).

5.1.2 Restrictions

To identify monetary policy shocks, we now introduce sign restrictions on the structural parameters, following [Arias, Caldara, and Rubio-Ramírez \(2019\)](#), and a correlation restriction, which uses the main feature of monetary policy (demand) shocks.¹¹ Specifically, we impose the following sign restrictions on the structural parameters.

Sign Restriction 1: *The contemporaneous reaction of the federal funds rate, the monetary policy instrument, to output growth and inflation is positive.*

Sign Restriction 1 means that the central bank increases the federal funds rate when output growth and inflation increase. In particular, we can write the equation for the federal funds rate as:

$$\begin{aligned} FFR_t = & \beta_g \Delta \log(GDP_t) + \beta_d \Delta \log(DEF_t) + \beta_c \Delta \log(CP_t) \\ & + \beta_{tr} \Delta \log(TR_t) + \beta_n \Delta \log(NBR_t) + \sigma_{\varepsilon_{FFR,t}} + \text{Other Terms}, \end{aligned}$$

where FFR , GDP , DEF , CP , TR , NBR , and $\varepsilon_{FFR,t}$ denote the federal funds rate, real GDP, the GDP deflator, the commodity price index, total reserves, nonborrowed reserves, and monetary policy shocks, respectively. The constant term and lag variables are included in Other Terms. **Sign Restriction 1** implies that β_g and β_d should be positive, which is consistent with monetary rules in New Keynesian DSGE models.

We complement **Sign Restriction 1** with a correlation restriction, which exploits the main feature of monetary policy (demand) shocks. Specifically, the correlation restriction is

¹⁰ Specifically, we use the first-difference specification in Section 6.3 of their paper.

¹¹ Note that, different from [Arias, Caldara, and Rubio-Ramírez \(2019\)](#), we do not impose zero restrictions. The results in which zero restrictions are imposed are shown in Online Appendix C.

Table 1: Conditional correlations between output and prices for monetary policy shocks

Restriction	Sign Restriction 1	Sign Restriction 1, Correlation Restriction 1
Posterior median	0.14	0.42
68% credible interval	[-0.68, 0.60]	[0.16, 0.71]

Note: The conditional correlations for each draw of the structural parameters are computed from the log-level IRFs of real GDP and the GDP deflator with $H_c = 60$ months.

summarized:

Correlation Restriction 1: *The conditional correlation between real GDP (output) and the GDP deflator (prices) for monetary policy shocks must be positive.*

Correlation Restriction 1 imposes an additional identifying assumption motivated by the main feature of monetary policy shocks. That is, output and prices should be positively correlated with respect to monetary policy shocks.

5.1.3 Results

Table 1 presents the conditional correlations between real GDP and the GDP deflator for monetary policy shocks, when only Sign Restriction 1 is used, and when both Sign Restriction 1 and Correlation Restriction 1 are used. For each draw of the structural parameters, we compute the correlation from the level IRFs with $H_c = 60$ months. The posterior median of the conditional correlation between real GDP and the GDP deflator for monetary policy shocks is 0.14, with a 68 percent credible interval of [-0.68, 0.60], when only Sign Restriction 1 is used. In contrast, when both restrictions are used, it is 0.42, with a 68 percent credible interval of [0.16, 0.71]. Despite the positive median under the sign restrictions, the credible interval spans a wide range of negative and positive correlations. Adding Correlation Restriction 1 excludes 42.5% of the draws admitted by the sign restrictions, ruling out comovement inconsistent with the demand shock interpretation.

Figure 1 displays IRFs of the six variables to a contractionary monetary policy shock with Sign Restriction 1 only and with both restrictions.¹² The lighter shaded area and dashed lines are the 68 percent point-wise credible sets and the median IRFs, respectively, using only Sign Restriction 1. The darker shaded area and solid lines show the equivalent quantities

¹² The IRFs are computed from 10,061 draws that satisfy Sign Restriction 1 and 5,787 draws that satisfy both Sign Restriction 1 and Correlation Restriction 1.

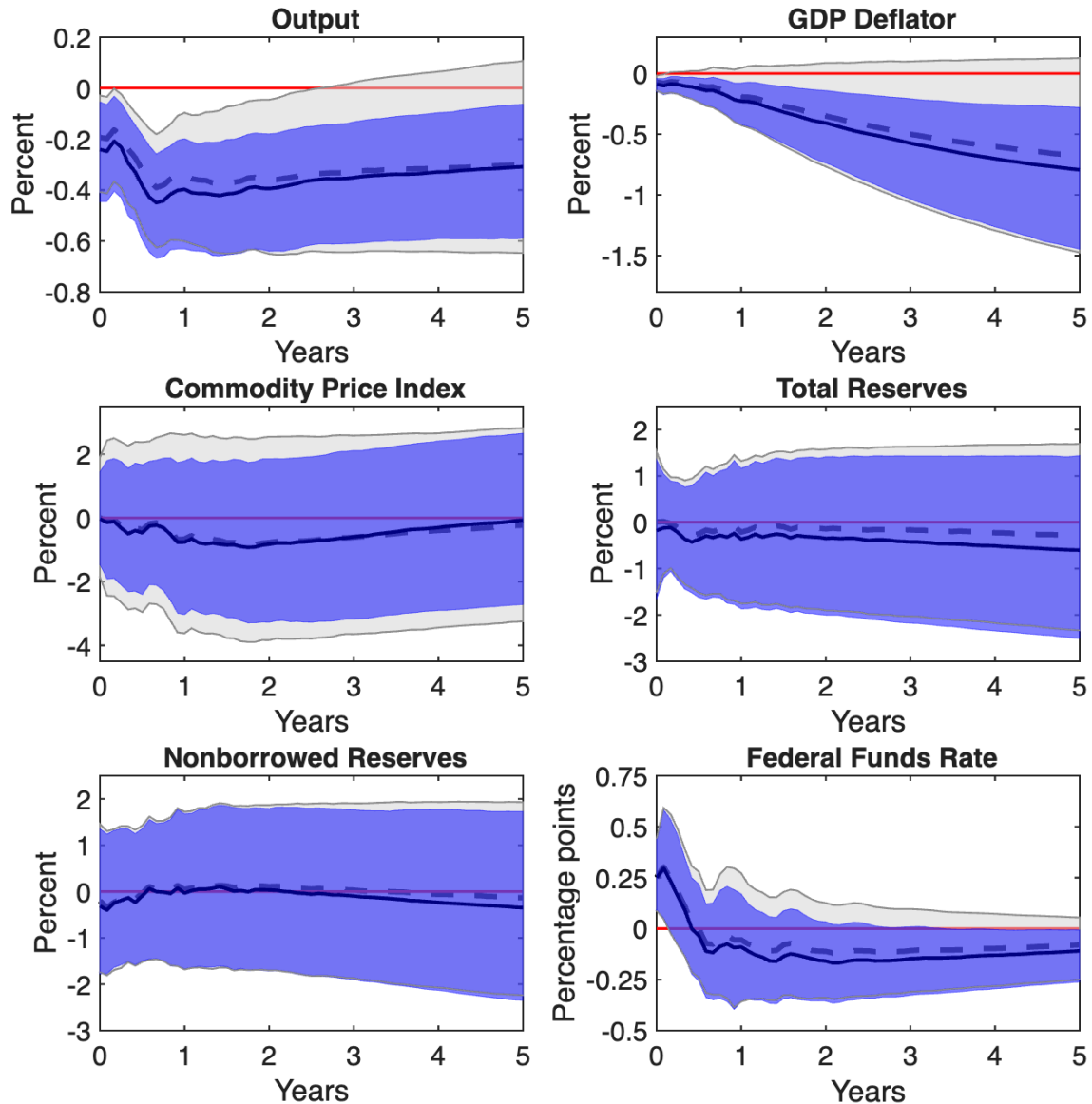


Figure 1: IRFs to a contractionary monetary policy shock

Notes: The lighter shaded area and dashed lines are the 68 percent point-wise credible sets and the median IRFs, respectively, using Sign Restriction 1. The darker shaded area and solid lines are the 68 percent point-wise credible sets and the median IRFs, respectively, using Sign Restriction 1 and Correlation Restriction 1. The IRFs of all variables, except for the federal funds rate, are accumulated to the log level.

when both Sign Restriction 1 and Correlation Restriction 1 are used. In both cases, the monetary policy shock is normalized to raise the federal funds rate by 25 basis points in period 0.

Overall, the IRFs in both cases have similar shapes. Specifically, a contractionary monetary policy shock increases immediately the federal funds rate. The shock leads to an immediate and persistent decline in output. Under the sign restrictions alone, however, its 68 percent credible set includes zero at longer horizons. Adding the correlation restriction keeps

this set below zero throughout the displayed horizon. The responses of commodity prices and total and nonborrowed reserves to the shock are close to zero. However, the response of prices is notably different in the two cases. With **Sign Restriction 1** only, the shock does not decrease prices significantly at most horizons, so the price puzzle cannot be ruled out. In contrast, the 68 percent credible set for the price response lies below zero when both **Sign Restriction 1** and **Correlation Restriction 1** are imposed. That is, the correlation restriction effectively eliminates the price puzzle.

Furthermore, the credible sets for all the IRFs with both restrictions are generally narrower than those with **Sign Restriction 1** only. In other words, the correlation restriction shrinks the credible sets, particularly for output and prices.

In a nutshell, the correlation restriction sharpens the identification of monetary policy shocks. Moreover, it makes the price puzzle disappear.

5.2 Oil supply shocks

A good example of supply shocks is oil supply shocks. An oil supply shock reduces world oil production, and hence the real oil price increases. Since oil is used in the production of almost every good, an increase in the real oil price raises production costs, which reduces aggregate supply. Accordingly, in response to the shock, output falls while prices rise. This comovement motivates a negative conditional correlation restriction. Therefore, we use this feature of oil supply shocks to identify oil supply shocks more precisely in this subsection.

5.2.1 Data and others

The SVAR model for oil supply shocks consists of four quarterly variables, as in [Baumeister and Peersman \(2013\)](#): US real GDP, US CPI, world oil production, and the real US refiners' acquisition cost of imported crude oil (the real oil price).¹³ The real oil price is defined as the nominal oil price deflated by US CPI. World oil production and the nominal US refiners' acquisition cost of imported crude oil are obtained from the US Energy Information Administration. The source of the seasonally adjusted US real GDP and CPI is the FRED. Following [Baumeister and Peersman \(2013\)](#), we estimate the SVAR using the first log difference of all the variables multiplied by 100. The sample period is from Q1 1974 to Q1 2011, which is the same as that in [Baumeister and Peersman \(2013\)](#). The SVAR includes 4 lags and a constant term, as in [Baumeister and Peersman \(2013\)](#).

¹³ Note that we do not include a variable for world real economic activity, such as the index of global real economic activity proposed by [Kilian \(2009\)](#), or the world industrial production index proposed by [Baumeister and Hamilton \(2019\)](#), because the focus of this subsection is oil supply shocks. Therefore, only oil supply shocks are explicitly identified, and there is no structural interpretation for the conglomerate of residual oil demand shocks.

5.2.2 Restrictions

Here, to identify oil supply shocks, we build on the oil-market sign restrictions of [Baumeister and Peersman \(2013\)](#) and add a restriction on the impact response of CPI. We also impose a correlation restriction based on the key feature of supply shocks. Specifically, we use the following sign restrictions on the IRFs.

Sign Restriction 2: *An oil supply shock reduces world oil production and increases the real oil price on impact and for the following four quarters. It also increases CPI on impact.*

The restrictions on world oil production and the real oil price are common in many studies, including [Baumeister and Peersman \(2013\)](#) and [Kim \(2020\)](#). We impose them on the log-level IRFs obtained by cumulating the responses from the first-difference VAR.

The additional CPI restriction is motivated by the direct pass-through of higher oil prices to the energy component of consumer prices. [Baumeister \(2023\)](#) reports an immediate increase in US headline inflation following an oil supply disruption. We therefore require CPI to rise in the quarter of the shock, but impose no signs on its subsequent responses or on the responses of GDP.

In addition, we also impose a correlation restriction based on the primary feature of oil supply shocks (supply shocks). Specifically, the correlation restriction is as follows:

Correlation Restriction 2: *The conditional correlation between real GDP (output) and CPI (prices) for oil supply shocks must be negative.*

Correlation Restriction 2 imposes an additional identifying assumption motivated by the primary feature of oil supply shocks, i.e., output and prices driven only by oil supply shocks should be negatively correlated.

5.2.3 Results

The conditional correlations between real GDP and CPI for oil supply shocks, when only **Sign Restriction 2** is used, and when both **Sign Restriction 2** and **Correlation Restriction 2** are used, are provided in Table 2. As in the previous subsection, the conditional correlations between real GDP and CPI for oil supply shocks are computed for each draw of the structural parameters from the level IRFs with $H_c = 20$ quarters. The posterior median of the conditional correlation between real GDP and CPI for oil supply shocks is -0.17, with a 68 percent credible interval of [-0.88, 0.61], when only **Sign Restriction 2** is imposed. In contrast, when both restrictions are imposed, it is -0.74, with a 68 percent credible interval

Table 2: Conditional correlation between output and prices for oil supply shocks

Restriction	Sign Restriction 2	Sign Restriction 2, Correlation Restriction 2
Posterior median	-0.17	-0.74
68% credible interval	[-0.88, 0.61]	[-0.93, -0.25]

Note: The correlations for each draw of the structural parameters are computed from the log-level IRFs of real GDP and CPI with $H_c = 20$ quarters.

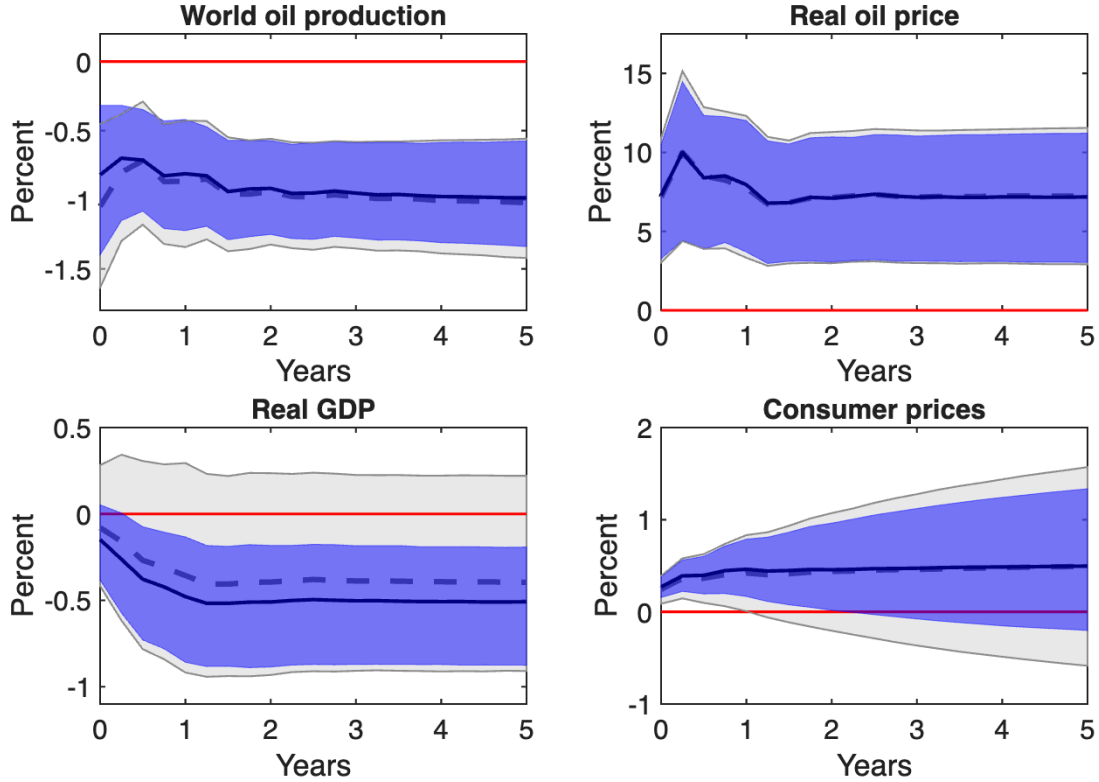


Figure 2: IRFs to an oil supply shock

Notes: The lighter shaded area and dashed lines are the 68 percent point-wise credible sets and the median IRFs, respectively, using Sign Restriction 2. The darker shaded area and solid lines are the 68 percent point-wise credible sets and the median IRFs, respectively, using Sign Restriction 2 and Correlation Restriction 2. The IRFs of all variables are accumulated to the log level.

of $[-0.93, -0.25]$. Under the sign restrictions alone, the median is negative, but the credible interval spans a wide range of negative and positive correlations. Imposing Correlation Restriction 2 removes 44.1% of the draws admitted by the sign restrictions because their implied comovement is inconsistent with the supply shock interpretation.

The IRFs of the four variables to an oil supply shock with and without Correlation Restriction 2 are shown in Figure 2.¹⁴ The lighter shaded area and dashed lines denote

¹⁴ The IRFs are computed from 9,000 draws that satisfy Sign Restriction 2 and 5,028 draws that satisfy

the 68 percent point-wise credible sets and the median IRFs with only **Sign Restriction 2**, respectively. The darker shaded area and solid lines show the equivalent quantities with **Sign Restriction 2** and **Correlation Restriction 2**. The oil supply shock is normalized to raise the real oil price by a maximum of 10 percent.

An oil supply shock decreases world oil production and increases the real oil price in both cases. The responses of the two variables are similar, although the 68 percent (point-wise) credible set for the IRF of world oil production with **Correlation Restriction 2** is smaller than that without the restriction. However, adding the correlation restriction makes the output decline more pronounced and narrows the credible sets for output and prices. With **Sign Restriction 2** only, the 68 percent credible set for output includes zero at every displayed horizon, whereas that for prices lies above zero on impact and for the following four quarters. Adding **Correlation Restriction 2** produces a larger median decline in output and a tighter credible set, which lies below zero from quarter 2 onward. Also, the 68 percent credible set for prices lies above zero on impact and for the following eight quarters. Moreover, the credible set for the IRF of prices is much narrower in the case with **Correlation Restriction 2** than in the case without the restriction.

In short, the correlation restriction helps identify oil supply shocks more sharply. Moreover, it provides clearer evidence of a persistent output decline and a more sustained price increase, which is in line with economic theory.

Finally, in Online Appendix D, we present the results when the narrative sign restrictions used in [Antolín-Díaz and Rubio-Ramírez \(2018\)](#), **Sign Restriction 2**, and **Correlation Restriction 2** are imposed to identify oil supply shocks. The results show that adding the correlation restriction further sharpens the contractionary output response even when the narrative sign restrictions are imposed.

5.3 Robust Bayesian inference

The IRF results in the applications, reported in Figures 1 and 2, use the conventional Bayesian summaries of posterior medians and 68 percent pointwise credible sets. These summaries are natural in our baseline analysis, but their interpretation is more delicate in a set-identified SVAR. The reason is that a given reduced form can generate multiple admissible structural paths through different rotations, and these paths have the same likelihood. The conventional practice based on a uniform prior over the rotation matrix can place some of these paths inside a reported credible set and others outside it, even though the likelihood does not distinguish among them. This point is emphasized by [Baumeister and Hamilton \(2024\)](#).

The issue matters for our applications because the SVARs also remain set-identified when

both **Sign Restriction 2** and **Correlation Restriction 2**.

the correlation restriction is imposed. One way to address this issue, emphasized by [Baumeister and Hamilton \(2024\)](#), is to specify and defend an explicit prior over the rotation matrix when ranking admissible structural paths. An alternative is the robust Bayesian approach of [Giacomini and Kitagawa \(2021\)](#). This approach avoids conditioning on a particular prior over rotations by summarizing the range of posterior mean IRFs attainable over admissible rotations. We implement it by computing the set of posterior means, obtained by taking the lower and upper bounds of the IRFs over the admissible rotations for each reduced-form posterior draw and then averaging these bounds over the reduced-form posterior distribution. Under this methodology, we compare the sign restrictions only with the sign and correlation restrictions.

The results are consistent with the main evidence: adding the correlation restriction rules out a sizable fraction of rotations that satisfy the sign restrictions and narrows the set of posterior means. The portions removed are responses that are less consistent with the shock interpretation, such as expansionary output or price responses to a contractionary monetary policy shock and positive output or negative consumer price responses to an oil supply shock. Online Appendix E provides the details.

6 Conclusion

This paper introduces the *correlation restriction* as an identifying restriction in Bayesian SVARs. This restriction is based on the key features of demand and supply shocks from economic theory. These features are that demand shocks move output and prices in the same direction, and that supply shocks move them in opposite directions. The correlation restriction constrains the structural parameters to ensure the conditional correlation between output and prices for demand (supply) shocks to be positive (negative). We also explain the Bayesian inference and the algorithm for sampling the structural parameters under both the sign restrictions and the correlation restriction.

The correlation restriction is applied to the SVARs for monetary policy shocks (demand shocks) and oil supply shocks (supply shocks). To identify structural monetary policy shocks, we impose a correlation restriction that the conditional correlation between output and prices for monetary policy shocks must be positive, whereas we impose a correlation restriction that the conditional correlation between the two variables for oil supply shocks must be negative. Estimation results show that the correlation restriction sharpens the identification of monetary policy shocks and enables the price puzzle to disappear. Furthermore, in the case of oil supply shocks, the correlation restriction helps shrink the uncertainty around the IRFs of all variables. In particular, under sign restrictions alone, the 68 percent credible set for output includes zero at every displayed horizon. Adding the correlation restriction provides

clearer evidence of a persistent output decline and a more sustained price increase, consistent with economic theory.

Finally, it should be noted that the approach in this paper has the advantage of restricting overall comovement without prescribing the signs of both IRFs at individual horizons. Specifically, demand and supply shocks can be distinguished by the comovement they generate. This economic distinction motivates the additional identifying assumption that the conditional correlation between output and prices for demand (supply) shocks is positive (negative). Therefore, many SVARs can easily make use of the correlation restriction that this paper introduces alongside existing identifying restrictions.

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Online Appendix for

“Restrictions on Conditional Correlations in SVARs”

Myunghyun Kim^{*} Sunho Lee[†]

Not for Publication

This appendix contains:

- A. Derivation of the conditional correlation
- B. Results under the strong correlation restriction
- C. Adding zero restrictions to the SVAR for monetary policy shocks
- D. Adding narrative sign restrictions to the SVAR for oil supply shocks
- E. Results under robust Bayesian inference
- F. Results under alternative correlation horizons

^{*} Department of Economics, Sungkyunkwan University, 25-2, Sungkyunkwan-ro, Jongno-gu, Seoul, 03063, Korea (e-mail: mhkim7812@gmail.com).

[†] Office of Economic Modeling & Policy Analysis, Bank of Korea, 39, Namdaemun-ro, Jung-gu, Seoul, 04531, Korea (e-mail: econ.preference@gmail.com).

A Derivation of the conditional correlation

This section derives the expected sample correlation between the paths of two variables in a linear system with time-invariant coefficients. Specifically, we first construct their paths using a new realization of the structural shock of interest in each period, with all other structural shocks set to zero.¹ For each resulting pair of paths, we then compute the sample correlation across time, holding the structural parameters Ψ fixed. Finally, we take its expectation over shock sequences. The derivation applies to both stationary and nonstationary variables.

Paths generated by a sequence of shocks Let $\mathbf{IR}_h^{(i,k)}(\Psi)$ and $\mathbf{IR}_h^{(j,k)}(\Psi)$ denote the IRFs of variables $y_{i,t}$ and $y_{j,t}$ to shock k . Normalize the initial calendar date to zero. Consider two paths for each variable, with the same initial history and deterministic terms. After date zero, one path is driven only by shocks of type k , and the other has no shocks. Let $\tilde{y}_{i,r}^k$ and $\tilde{y}_{j,r}^k$ denote the deviations of the two variables from their respective paths without shocks. By linearity, the common initial-state and deterministic components cancel. Hence, $\tilde{y}_{i,0}^k = \tilde{y}_{j,0}^k = 0$ by construction. This zero initialization applies to the shock contributions, not to the observed variables or their lagged values.

By linearity, adding the responses to shocks occurring at dates $1, \dots, r$ gives, for $r \geq 1$,

$$\begin{aligned} \tilde{y}_{i,r}^k &= \sum_{s=1}^r \mathbf{IR}_{r-s}^{(i,k)}(\Psi) \varepsilon_s^k \\ \text{and } \tilde{y}_{j,r}^k &= \sum_{s=1}^r \mathbf{IR}_{r-s}^{(j,k)}(\Psi) \varepsilon_s^k. \end{aligned} \tag{A.1}$$

Each term is the shock at date s multiplied by the corresponding IRF at horizon $r - s$. This construction involves only finitely many shocks and does not rely on the stationary Wold decomposition theorem. The expressions therefore remain valid when shocks have permanent effects on levels or the variables are otherwise nonstationary.

We measure the correlation over $H_c \geq 2$ observations at dates $1, \dots, H_c$. Let the shock sequence be $\boldsymbol{\varepsilon}^k = (\varepsilon_1^k, \dots, \varepsilon_{H_c}^k)'$, with $\boldsymbol{\varepsilon}^k \sim N(\mathbf{0}, \mathbf{I}_{H_c})$. Stack the observations as

$$\tilde{\mathbf{y}}_i^k = (\tilde{y}_{i,1}^k, \dots, \tilde{y}_{i,H_c}^k)',$$

and define $\tilde{\mathbf{y}}_j^k$ analogously. Define the $H_c \times H_c$ matrix $\mathbf{I}_i^k$, with row index $r = 1, \dots, H_c$ and

¹ The term ‘conditional’ refers to conditioning on all other structural shocks being zero. It does not refer to conditioning on the information available from past observations.

column index $s = 1, \dots, H_c$, by

$$(\mathbf{L}_i^k)_{r,s} = \begin{cases} \mathbf{I}\mathbf{R}_{r-s}^{(i,k)}(\Psi), & s \leq r, \\ 0, & s > r, \end{cases} \quad (\text{A.2})$$

and define $\mathbf{L}_j^k$ analogously. The two matrices are lower triangular. Equation (A.1) then implies

$$\tilde{\mathbf{y}}_i^k = \mathbf{L}_i^k \boldsymbol{\varepsilon}^k, \quad \tilde{\mathbf{y}}_j^k = \mathbf{L}_j^k \boldsymbol{\varepsilon}^k. \quad (\text{A.3})$$

Sample correlation and its expectation Let $\mathbf{M}_{H_c} = \mathbf{I}_{H_c} - \mathbf{1}_{H_c} \mathbf{1}_{H_c}' / H_c$, where $\mathbf{1}_{H_c}$ is the H_c -dimensional vector of ones. Multiplication by $\mathbf{M}_{H_c}$ removes each path's sample mean. Define

$$\mathbf{K}_{ii}^k = (\mathbf{L}_i^k)' \mathbf{M}_{H_c} \mathbf{L}_i^k, \quad (\text{A.4})$$

$$\mathbf{K}_{jj}^k = (\mathbf{L}_j^k)' \mathbf{M}_{H_c} \mathbf{L}_j^k, \quad (\text{A.5})$$

$$\text{and } \mathbf{K}_{ij}^k = \frac{1}{2} [(\mathbf{L}_i^k)' \mathbf{M}_{H_c} \mathbf{L}_j^k + (\mathbf{L}_j^k)' \mathbf{M}_{H_c} \mathbf{L}_i^k]. \quad (\text{A.6})$$

The first two matrices are positive semidefinite, and all three are symmetric. The sample covariance and variances of the paths are

$$s_{ij}^k = \frac{(\tilde{\mathbf{y}}_i^k)' \mathbf{M}_{H_c} \tilde{\mathbf{y}}_j^k}{H_c - 1} = \frac{(\boldsymbol{\varepsilon}^k)' \mathbf{K}_{ij}^k \boldsymbol{\varepsilon}^k}{H_c - 1}, \quad (\text{A.7})$$

$$s_{ii}^k = \frac{(\boldsymbol{\varepsilon}^k)' \mathbf{K}_{ii}^k \boldsymbol{\varepsilon}^k}{H_c - 1}, \quad (\text{A.8})$$

$$\text{and } s_{jj}^k = \frac{(\boldsymbol{\varepsilon}^k)' \mathbf{K}_{jj}^k \boldsymbol{\varepsilon}^k}{H_c - 1}. \quad (\text{A.9})$$

Thus, the sample correlation is

$$\hat{\rho}_{i,j}^k(\boldsymbol{\varepsilon}^k; H_c) = \frac{s_{ij}^k}{\sqrt{s_{ii}^k s_{jj}^k}} = \frac{(\boldsymbol{\varepsilon}^k)' \mathbf{K}_{ij}^k \boldsymbol{\varepsilon}^k}{\sqrt{(\boldsymbol{\varepsilon}^k)' \mathbf{K}_{ii}^k \boldsymbol{\varepsilon}^k} \sqrt{(\boldsymbol{\varepsilon}^k)' \mathbf{K}_{jj}^k \boldsymbol{\varepsilon}^k}}. \quad (\text{A.10})$$

We assume that both sample variances are positive with probability one. The expected sample correlation is

$$\rho_{i,j}^k(\Psi; H_c) = E_{\boldsymbol{\varepsilon}^k} [\hat{\rho}_{i,j}^k(\boldsymbol{\varepsilon}^k; H_c)]. \quad (\text{A.11})$$

Analytical evaluation Following the proof of Proposition 1 in [Ernst et al. \(2025\)](#), we derive the first moment of the finite sample correlation in Equation (A.10). For $a, b > 0$ and

$c^2 < ab$, define

$$\begin{aligned}
\varphi^k(a, c, b) &= E_{\boldsymbol{\varepsilon}^k} \left[\exp \left\{ -\frac{1}{2} (\boldsymbol{\varepsilon}^k)' (a \mathbf{K}_{ii}^k + 2c \mathbf{K}_{ij}^k + b \mathbf{K}_{jj}^k) \boldsymbol{\varepsilon}^k \right\} \right] \\
&= \frac{1}{(2\pi)^{H_c/2}} \int_{\mathbb{R}^{H_c}} \exp \left\{ -\frac{1}{2} \mathbf{z}' (\mathbf{I}_{H_c} + a \mathbf{K}_{ii}^k + 2c \mathbf{K}_{ij}^k + b \mathbf{K}_{jj}^k) \mathbf{z} \right\} d\mathbf{z} \\
&= [\det(\mathbf{I}_{H_c} + a \mathbf{K}_{ii}^k + 2c \mathbf{K}_{ij}^k + b \mathbf{K}_{jj}^k)]^{-1/2}, \tag{A.12}
\end{aligned}$$

where $\boldsymbol{\varepsilon}^k \sim N(\mathbf{0}, \mathbf{I}_{H_c})$. The final equality follows from the Gaussian normalizing constant.

For $x > 0$, the gamma integral gives

$$x^{-1/2} = \frac{1}{\sqrt{2\pi}} \int_0^\infty a^{-1/2} e^{-ax/2} da.$$

Applying this identity to both denominator terms in Equation (A.10) and interchanging expectation and integration gives

$$\begin{aligned}
\rho_{i,j}^k(\boldsymbol{\Psi}; H_c) &= \frac{1}{2\pi} \int_0^\infty \int_0^\infty \frac{E_{\boldsymbol{\varepsilon}^k} \left[(\boldsymbol{\varepsilon}^k)' \mathbf{K}_{ij}^k \boldsymbol{\varepsilon}^k e^{-\frac{1}{2} (\boldsymbol{\varepsilon}^k)' (a \mathbf{K}_{ii}^k + b \mathbf{K}_{jj}^k) \boldsymbol{\varepsilon}^k} \right]}{\sqrt{ab}} da db, \\
&= -\frac{1}{2\pi} \int_0^\infty \int_0^\infty \frac{1}{\sqrt{ab}} \frac{\partial \varphi^k}{\partial c}(a, 0, b) da db. \tag{A.13}
\end{aligned}$$

For the second equality, differentiating φ^k with respect to c introduces the factor $-(\boldsymbol{\varepsilon}^k)' \mathbf{K}_{ij}^k \boldsymbol{\varepsilon}^k$ inside the expectation. Evaluating at $c = 0$ then removes the term multiplied by c from the exponent. The interchange of expectation and integration is justified by Fubini's theorem, since absolute integrability follows from $E_{\boldsymbol{\varepsilon}^k}[\|\hat{\rho}_{i,j}^k(\boldsymbol{\varepsilon}^k; H_c)\|] \leq 1$.

To express Equation (A.13) in the form used in the main text, set $a = u^2$ and $b = v^2$ and define

$$\mathbf{D}^k(u, v) = \mathbf{I}_{H_c} + u^2 \mathbf{K}_{ii}^k + v^2 \mathbf{K}_{jj}^k. \tag{A.14}$$

The variables u and v are auxiliary integration variables, not time indices. Differentiating $\varphi^k(a, c, b)$ with respect to c and setting $c = 0$ gives

$$\frac{\partial \varphi^k}{\partial c}(u^2, 0, v^2) = -\frac{\text{tr}[\mathbf{K}_{ij}^k \mathbf{D}^k(u, v)^{-1}]}{\sqrt{\det \mathbf{D}^k(u, v)}}. \tag{A.15}$$

Note that $\mathbf{D}^k(u, v)$ is positive definite because $\mathbf{K}_{ii}^k$ and $\mathbf{K}_{jj}^k$ are positive semidefinite. Its inverse therefore exists and its determinant is strictly positive.

Substituting into Equation (A.13) and using $da db/\sqrt{ab} = 4 du dv$ yields

$$\rho_{i,j}^k(\Psi; H_c) = \frac{2}{\pi} \int_0^\infty \int_0^\infty \frac{\text{tr}[\mathbf{K}_{ij}^k \mathbf{D}^k(u, v)^{-1}]}{\sqrt{\det \mathbf{D}^k(u, v)}} du dv. \quad (\text{A.16})$$

Numerical evaluation We evaluate Equation (A.16) in polar coordinates, setting $u = r \cos \theta$ and $v = r \sin \theta$, with $r \geq 0$ and $0 \leq \theta \leq \pi/2$. For each θ , let $\lambda_m^k(\theta)$ and $\mathbf{v}_m^k(\theta)$, $m = 1, \dots, H_c$, denote the m th eigenvalue and its corresponding eigenvector, respectively, of

$$\mathbf{G}^k(\theta) = \cos^2 \theta \mathbf{K}_{ii}^k + \sin^2 \theta \mathbf{K}_{jj}^k.$$

The eigenvectors are chosen to be orthonormal. Define $c_m^k(\theta) = (\mathbf{v}_m^k(\theta))' \mathbf{K}_{ij}^k \mathbf{v}_m^k(\theta)$. Equation (A.16) becomes

$$\rho_{i,j}^k(\Psi; H_c) = \frac{2}{\pi} \int_0^{\pi/2} \int_0^\infty r \frac{\sum_{m=1}^{H_c} \frac{c_m^k(\theta)}{1 + r^2 \lambda_m^k(\theta)}}{\sqrt{\prod_{m=1}^{H_c} (1 + r^2 \lambda_m^k(\theta))}} dr d\theta. \quad (\text{A.17})$$

This representation requires only one eigendecomposition per angle, avoiding repeated matrix inversions as r varies.

The radial and angular integrals are evaluated using adaptive Gauss–Kronrod quadrature with an embedded 7/15-point rule. For the angular integral, we apply the change of variables $\theta = (\pi/4)(1 + 3t/2 - t^3/2)$, $-1 \leq t \leq 1$, before numerical integration.

B Results under the strong correlation restriction

Here, we consider a strong correlation restriction. Specifically, instead of **Correlation Restriction 1**, we impose the following correlation restriction for the SVAR for monetary policy shocks:

Correlation Restriction 1.1: *The conditional correlation between real GDP (output) and the GDP deflator (prices) for monetary policy shocks must be larger than 0.2.*

Figure B.1 shows the IRFs of the six variables to a contractionary monetary policy shock with **Sign Restriction 1** and with both **Sign Restriction 1** and **Correlation Restriction 1.1**.² The lighter shaded area and dashed lines are the 68 percent point-wise credible sets

² The IRFs are computed from 10,061 draws that satisfy **Sign Restriction 1** and 4,616 draws that satisfy **Sign Restriction 1** and **Correlation Restriction 1.1**.

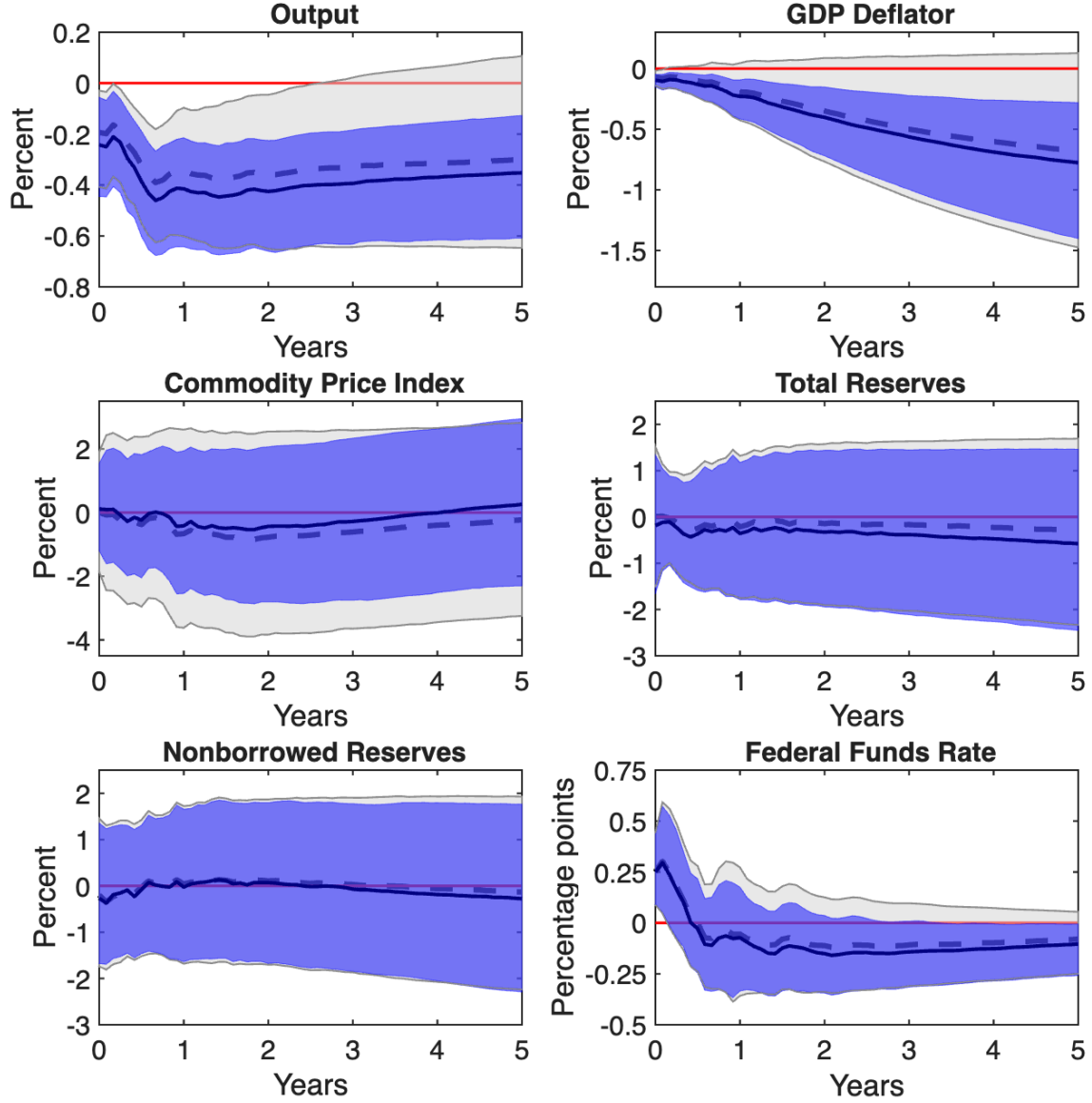


Figure B.1: IRFs to a contractionary monetary policy shock (**Correlation Restriction 1.1**)

Notes: The lighter shaded area and dashed lines are the 68 percent point-wise credible sets and the median IRFs, respectively, using **Sign Restriction 1**. The darker shaded area and solid lines are the 68 percent point-wise credible sets and the median IRFs, respectively, using **Sign Restriction 1** and **Correlation Restriction 1.1**. The IRFs of all variables, except for the federal funds rate, are accumulated to the log level.

and the median IRFs, respectively, using only **Sign Restriction 1**. The darker shaded area and solid lines show the equivalent quantities when **Sign Restriction 1** and **Correlation Restriction 1.1** are used. In both cases, the monetary policy shock is normalized to raise the federal funds rate by 25 basis points in period 0. The conditional correlations between real GDP and the GDP deflator for monetary policy shocks are computed for each draw of the structural parameters from the level IRFs with $H_c = 60$ months, as in the main text. The posterior median of the correlation is 0.14, with a 68 percent credible interval of $[-0.68, 0.60]$, when only **Sign Restriction 1** is used. In contrast, when **Sign Restriction 1** and

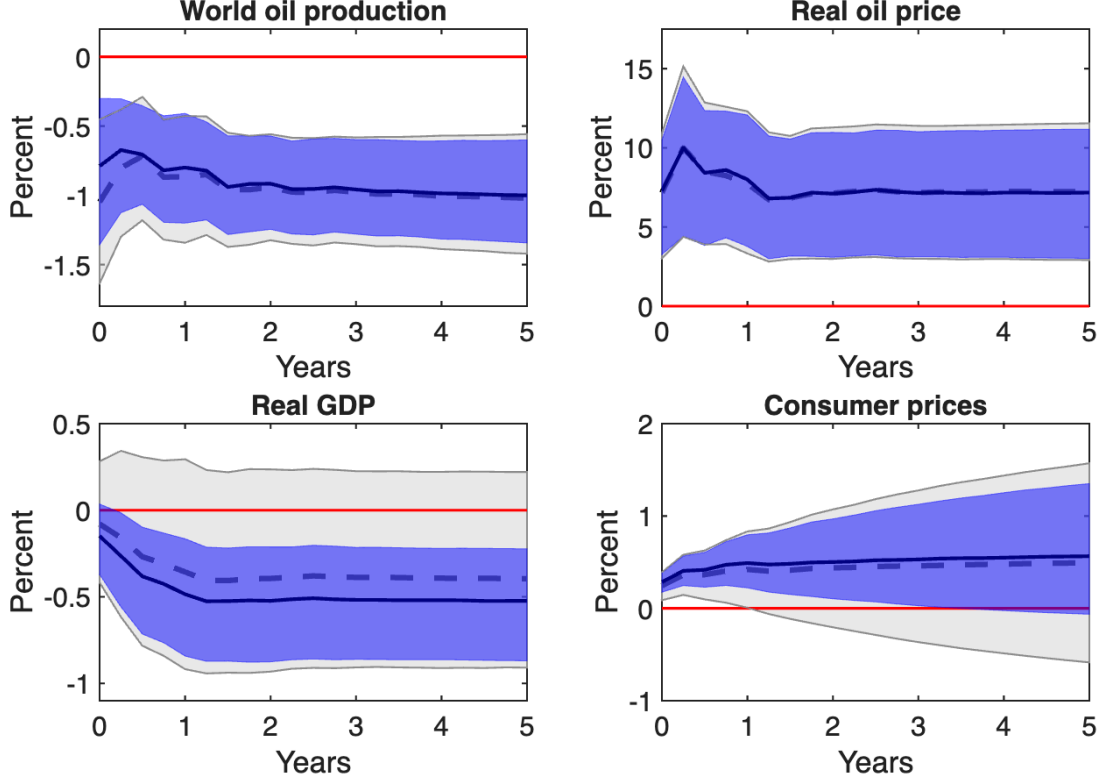


Figure B.2: IRFs to an oil supply shock (Correlation Restriction 2.1)

Notes: The lighter shaded area and dashed lines are the 68 percent point-wise credible sets and the median IRFs, respectively, using Sign Restriction 2. The darker shaded area and solid lines are the 68 percent point-wise credible sets and the median IRFs, respectively, using Sign Restriction 2 and Correlation Restriction 2.1. The IRFs of all variables are accumulated to the log level.

Correlation Restriction 1.1 are used, it is 0.50, with a 68 percent credible interval of $[0.30, 0.74]$.

The IRFs of all six variables are almost identical to those in Figure 1 in the main text. Therefore, the correlation restriction, regardless of whether it is weak or strong, sharpens the identification of monetary policy shocks and helps the price puzzle to disappear.

Then, we consider a strong correlation restriction in the SVAR for oil supply shocks. Instead of Correlation Restriction 2, we impose the following correlation restriction:

Correlation Restriction 2.1: *The conditional correlation between real GDP (output) and CPI (prices) for oil supply shocks must be smaller than -0.2 .*

Figure B.2 presents the IRFs of the four variables to an oil supply shock with Sign Restriction 2 and with both Sign Restriction 2 and Correlation Restriction 2.1.³ The lighter shaded area and dashed lines are the 68 percent point-wise credible sets and the median IRFs with Sign Restriction 2. The darker shaded area and solid lines show the equivalent

³ The IRFs are computed from 9,000 draws that satisfy Sign Restriction 2 and 4,360 draws that satisfy Sign Restriction 2 and Correlation Restriction 2.1.

quantities with **Sign Restriction 2** and **Correlation Restriction 2.1**. The oil supply shock is normalized to raise the real oil price by a maximum of 10 percent. The conditional correlations between real GDP and CPI for oil supply shocks are computed for each draw of the structural parameters from the level IRFs with $H_c = 20$ quarters, as in the main text. The posterior median of the correlation is -0.17, with a 68 percent credible interval of [-0.88, 0.61], when only **Sign Restriction 2** is imposed. In contrast, when **Sign Restriction 2** and **Correlation Restriction 2.1** are imposed, it is -0.79, with a 68 percent credible interval of [-0.94, -0.43].

The overall pattern of the IRFs in Figure B.2 is similar to that in Figure 2 in the main text. Under the strong correlation restriction, the 68 percent credible set for output lies below zero from quarter 1 onward, and that for prices lies above zero on impact and for the following fourteen quarters. Hence, regardless of whether it is strong or weak, the correlation restriction sharpens the evidence of an output decline and a more sustained price increase.

C Adding zero restrictions to the SVAR for monetary policy shocks

In this appendix, we add the zero restrictions used in [Arias, Caldara, and Rubio-Ramírez \(2019\)](#) to the SVAR for monetary policy shocks in the main text. Specifically, the following zero restrictions are imposed:

Zero Restriction 1: *The federal funds rate only reacts contemporaneously to output growth, inflation, and commodity price growth.*

Zero Restriction 1 implies that the federal funds rate does not react to total and nonborrowed reserves, which means that the two contemporaneous reserves are not included in the equation for the federal funds rate.

Figure C shows the IRFs of the six variables to a contractionary monetary policy shock with **Sign Restriction 1** and **Zero Restriction 1**, and with the two restrictions and **Correlation Restriction 1**.⁴ The lighter shaded area and dashed lines are the 68 percent point-wise credible sets and the median IRFs, respectively, using only **Sign Restriction 1** and **Zero Restriction 1**. The darker shaded area and solid lines show the equivalent quantities when **Sign Restriction 1**, **Zero Restriction 1**, and **Correlation Restriction 1** are used. In both cases, the monetary policy shock is normalized to raise the federal funds rate by 25 basis points in period 0. The conditional correlations between real GDP and

⁴ The IRFs are computed from 9,449 draws that satisfy **Sign Restriction 1** and **Zero Restriction 1** and 5,697 draws that satisfy the two restrictions and **Correlation Restriction 1**.

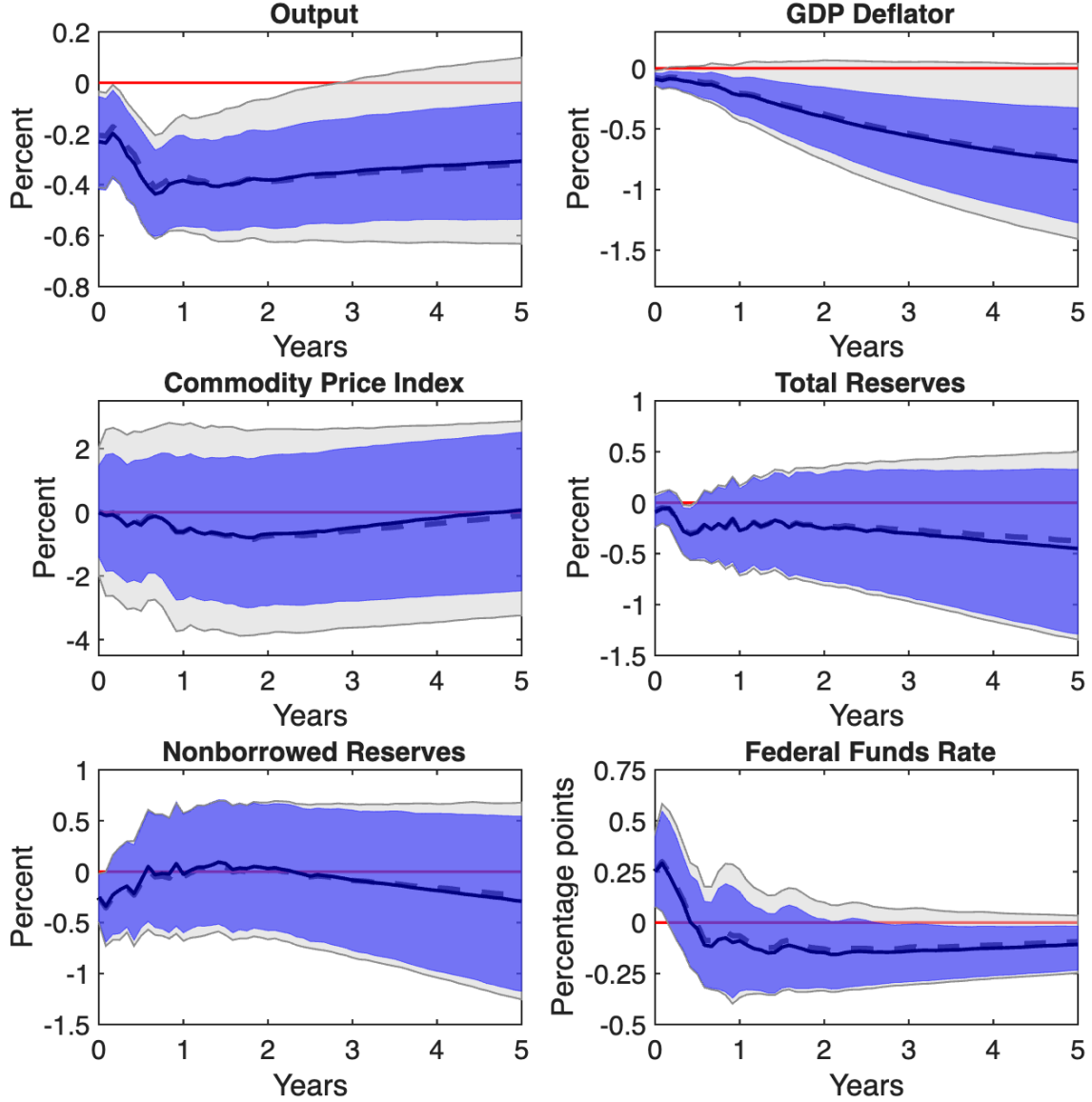


Figure C: IRFs to a contractionary monetary policy shock (Zero Restriction 1)

Notes: The lighter shaded area and dashed lines are the 68 percent point-wise credible sets and the median IRFs, respectively, using **Sign Restriction 1** and **Zero Restriction 1**. The darker shaded area and solid lines are the 68 percent point-wise credible sets and the median IRFs, respectively, using **Sign Restriction 1**, **Zero Restriction 1**, and **Correlation Restriction 1**. The IRFs of all variables, except for the federal funds rate, are accumulated to the log level.

the GDP deflator for monetary policy shocks are computed for each draw of the structural parameters from the level IRFs with $H_c = 60$ months, as in the main text. The posterior median of the correlation is 0.19, with a 68 percent credible interval of $[-0.67, 0.62]$, when **Sign Restriction 1** and **Zero Restriction 1** are used. In contrast, when **Sign Restriction 1**, **Zero Restriction 1**, and **Correlation Restriction 1** are used, it is 0.45, with a 68 percent credible interval of $[0.18, 0.72]$.

The overall contour of the IRFs is very similar to those in Figure 1 in the main text. As Figure C clearly shows, the correlation restriction sharpens the identification of monetary

policy shocks and allows the price puzzle to disappear, even when Zero Restriction 1 is additionally imposed.

D Adding narrative sign restrictions to the SVAR for oil supply shocks

In this appendix, we add the narrative sign restrictions used in [Antolín-Díaz and Rubio-Ramírez \(2018\)](#) to the SVAR for oil supply shocks in the main text. Specifically, the following narrative sign restrictions are imposed:

Narrative Sign Restriction 1: *The oil supply shock must take positive values in Q1 1979, Q4 1980, Q3 1990, Q4 2002, Q1 2003, and Q1 2011.*

Narrative Sign Restriction 1 is slightly different from those in [Antolín-Díaz and Rubio-Ramírez \(2018\)](#). This is mainly because a monthly SVAR is estimated in [Antolín-Díaz and Rubio-Ramírez \(2018\)](#), while we estimate a quarterly SVAR. The associated events are the Iranian Revolution (December 1978–January 1979), the Iran-Iraq War (September–October 1980), the Persian Gulf War (August 1990), the Venezuela oil strike of December 2002, the start of the Iraq War (March 2003), and the Libyan Civil War (February 2011). For each event, we impose the restriction in the quarter containing the last month of the period specified in [Antolín-Díaz and Rubio-Ramírez \(2018\)](#). For example, for the Iranian Revolution, which occurs December 1978–January 1979, we impose a narrative sign restriction in Q1 1979. Note also that since we identify only oil supply shocks, we do not impose narrative sign restrictions on the contribution of a certain shock in the quarter when the relevant event occurs. That is, we do not impose restrictions on the historical decomposition around relevant events.

Figure D presents the IRFs of the four variables to an oil supply shock with Sign Restriction 2 and Narrative Sign Restriction 1, and with the two restrictions and Correlation Restriction 2.⁵ The lighter shaded area and dashed lines are the 68 percent point-wise credible sets and the median IRFs with Sign Restriction 2 and Narrative Sign Restriction 1. The darker shaded area and solid lines show the equivalent quantities with Sign Restriction 2, Narrative Sign Restriction 1, and Correlation Restriction 2. The oil supply shock is normalized to raise the real oil price by a maximum of 10 percent. The conditional correlations between real GDP and CPI for oil supply shocks are computed for each draw of the structural parameters from the level IRFs with $H_c = 20$ quarters, as in the main text. The posterior median of the correlation is -0.81, with a 68 percent credible

⁵ The IRFs are computed from 6,842 draws that satisfy Sign Restriction 2 and Narrative Sign Restriction 1, and 6,158 draws that satisfy Sign Restriction 2, Narrative Sign Restriction 1, and Correlation Restriction 2.

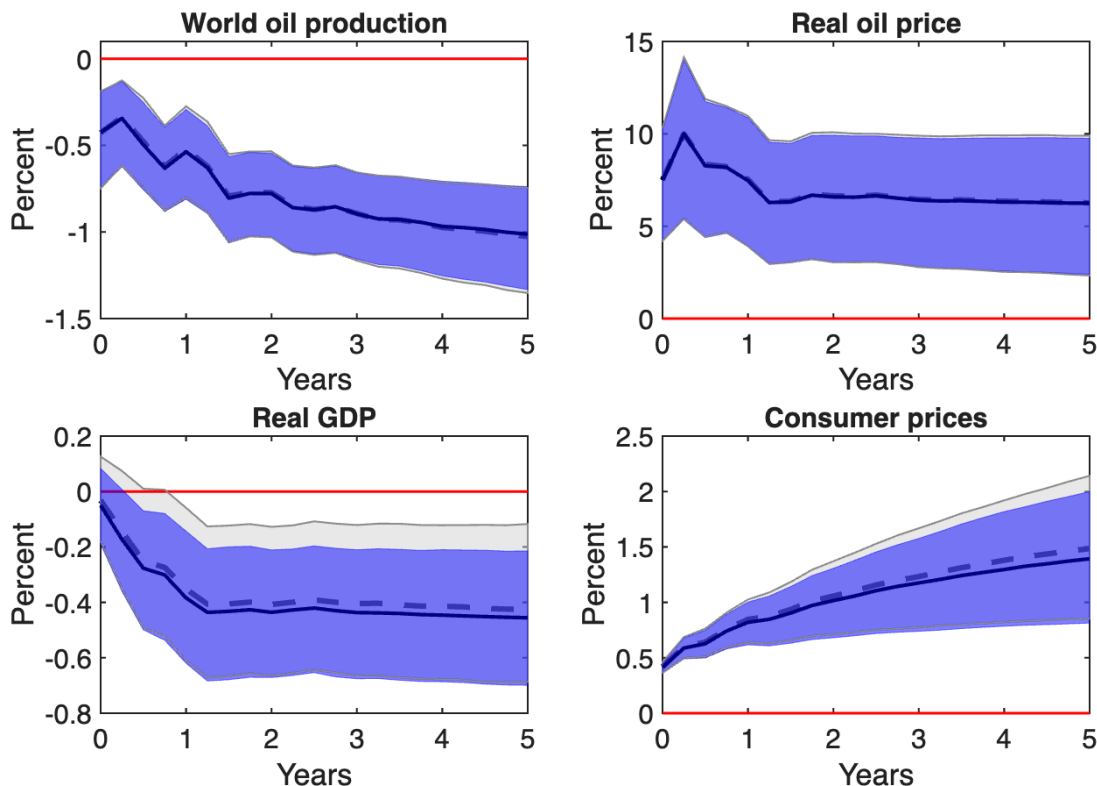


Figure D: IRFs to an oil supply shock (Narrative Sign Restriction 1)

Notes: The lighter shaded area and dashed lines are the 68 percent point-wise credible sets and the median IRFs, respectively, using Sign Restriction 2 and Narrative Sign Restriction 1. The darker shaded area and solid lines are the 68 percent point-wise credible sets and the median IRFs, respectively, using Sign Restriction 2, Narrative Sign Restriction 1, and Correlation Restriction 2. The IRFs of all variables are accumulated to the log level.

interval of $[-0.92, -0.35]$, when Sign Restriction 2 and Narrative Sign Restriction 1 are imposed. In contrast, when Sign Restriction 2, Narrative Sign Restriction 1, and Correlation Restriction 2 are imposed, it is -0.83 , with a 68 percent credible interval of $[-0.92, -0.59]$.

The main difference between the two cases concerns output. With the sign and narrative sign restrictions alone, the 68 percent credible set for output lies below zero from quarter 4 onward. Adding Correlation Restriction 2 produces a larger median decline in output and a tighter credible set, which lies below zero from quarter 2 onward. The price responses remain similar, with credible sets above zero throughout the displayed horizon in both cases. Thus, the correlation restriction further sharpens the contractionary output response even when the narrative sign restrictions are imposed.

E Results under robust Bayesian inference

In set-identified SVARs, a given reduced form can be associated with multiple structural paths that satisfy the identifying restrictions and achieve the same likelihood value. As emphasized

by [Baumeister and Hamilton \(2024\)](#), the conventional practice based on a uniform prior over the rotation matrix can place some admissible paths inside a reported credible set and others outside it, even though the likelihood does not distinguish among them. One way to address this issue is to specify and defend an explicit prior over the rotation matrix when ranking admissible structural paths. Another way is to use the robust Bayesian approach of [Giacomini and Kitagawa \(2021\)](#), which avoids conditioning on a particular prior over rotations. Since the SVARs in this paper remain set-identified after imposing the correlation restriction, it is useful to examine, within this framework, whether the correlation restriction adds identifying content beyond the sign restrictions. We therefore compare the results under the sign restrictions only with those under the sign and correlation restrictions.

For each reduced-form posterior draw $(\mathbf{B}, \Sigma)$, let $\mathcal{Q}_{\mathcal{R}}(\mathbf{B}, \Sigma)$ be the set of rotations that satisfy a restriction set $\mathcal{R}$, such as sign or correlation restrictions. For the shock of interest k , variable i , and horizon h , define

$$\ell_{i,h}^{\mathcal{R}}(\mathbf{B}, \Sigma) = \inf_{\mathbf{Q} \in \mathcal{Q}_{\mathcal{R}}(\mathbf{B}, \Sigma)} \mathbf{IR}_h^{(i,k)}(\mathbf{B}, \Sigma, \mathbf{Q}) \quad (\text{E.1})$$

$$\text{and } u_{i,h}^{\mathcal{R}}(\mathbf{B}, \Sigma) = \sup_{\mathbf{Q} \in \mathcal{Q}_{\mathcal{R}}(\mathbf{B}, \Sigma)} \mathbf{IR}_h^{(i,k)}(\mathbf{B}, \Sigma, \mathbf{Q}). \quad (\text{E.2})$$

The set of posterior means is

$$\left[E_{\mathbf{B}, \Sigma | Y} [\ell_{i,h}^{\mathcal{R}}(\mathbf{B}, \Sigma)], E_{\mathbf{B}, \Sigma | Y} [u_{i,h}^{\mathcal{R}}(\mathbf{B}, \Sigma)] \right]. \quad (\text{E.3})$$

This interval gives the lower and upper bounds on the posterior mean IRF that can be attained by rotations satisfying the restriction set $\mathcal{R}$, and it is the object reported in this section.

We approximate the bounds by simulation. For each reduced-form draw, we draw a fixed number of Haar rotations and retain those that satisfy the restrictions. We then compute the minimum and maximum IRFs over the retained rotations, and average these bounds across reduced-form draws. This follows the simulation approach in [Giacomini and Kitagawa \(2021\)](#).

For the monetary policy application, we retain 3,000 reduced-form draws and use 10,000 rotations for each draw. Among the rotations satisfying the sign restrictions, 41.8% do not satisfy the correlation restriction. For the oil supply application, we process 3,000 reduced-form draws and use 150,000 rotations for each draw. Among the rotations satisfying the sign restrictions, 45.7% do not satisfy the correlation restriction. These results show that the correlation restriction substantially reduces the set of rotations admitted by the sign restrictions.

Figure E.1 reports the results for monetary policy shocks. Adding the correlation restriction narrows the set of posterior means. The reduction is clear for output and the GDP

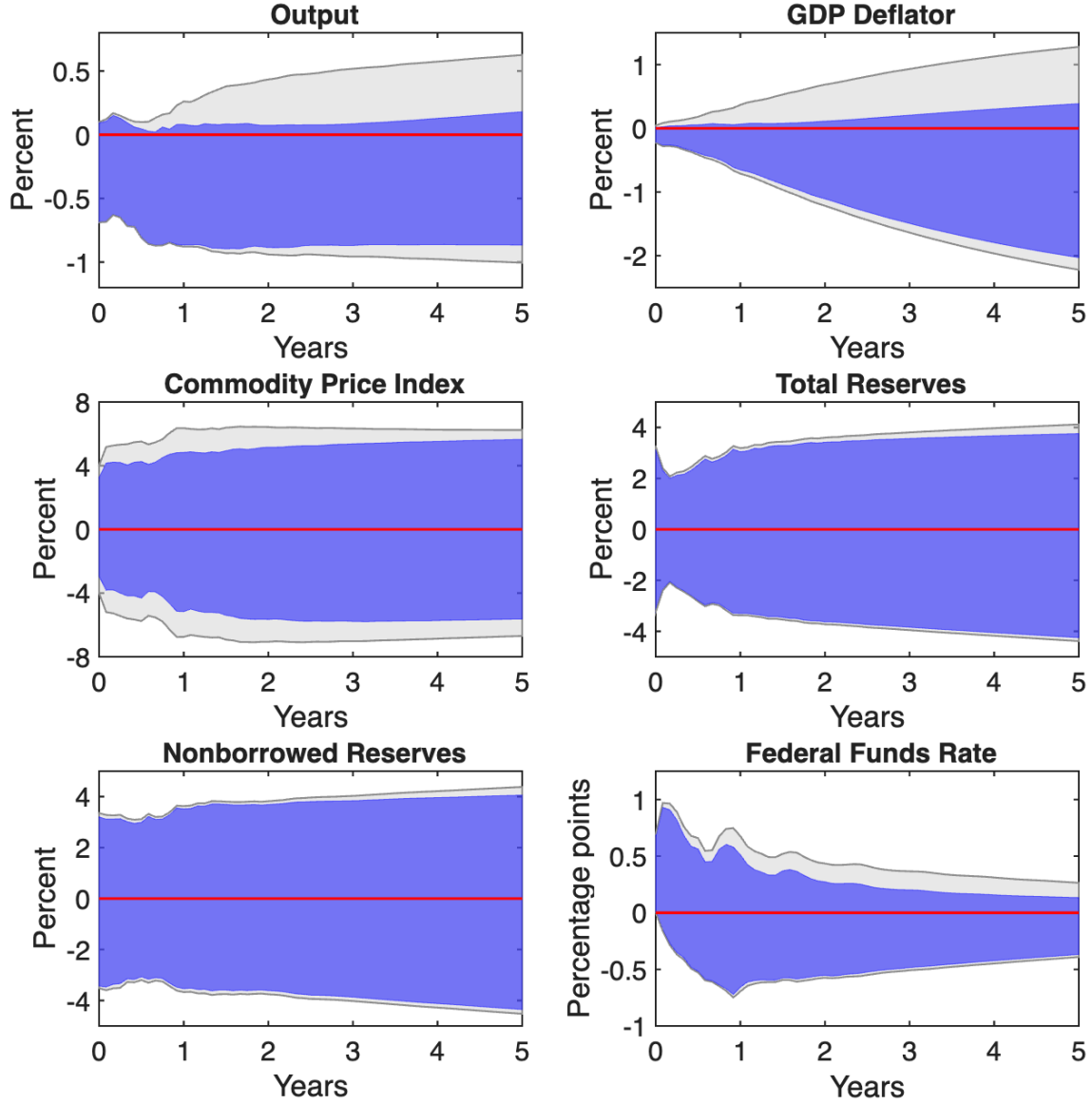


Figure E.1: Set of posterior means for a contractionary monetary policy shock

Notes: The lighter shaded area is the set of posterior means using **Sign Restriction 1**. The darker blue shaded area is the set of posterior means using **Sign Restriction 1** and **Correlation Restriction 1**. The IRFs of all variables, except for the federal funds rate, are accumulated to the log level.

deflator. For output, the correlation restriction removes much of the positive response allowed under the sign restrictions only, making the contractionary output response more tightly identified. For the GDP deflator, it also removes much of the positive response, indicating that the price-puzzle pattern is substantially weakened.

Figure E.2 reports the results for oil supply shocks. Adding the correlation restriction narrows the set of posterior means for output and consumer prices. The average widths over horizons 0–20 fall by 29.1% for output and 26.3% for consumer prices. Under the sign restrictions only, the sets allow substantial positive output responses and negative price re-

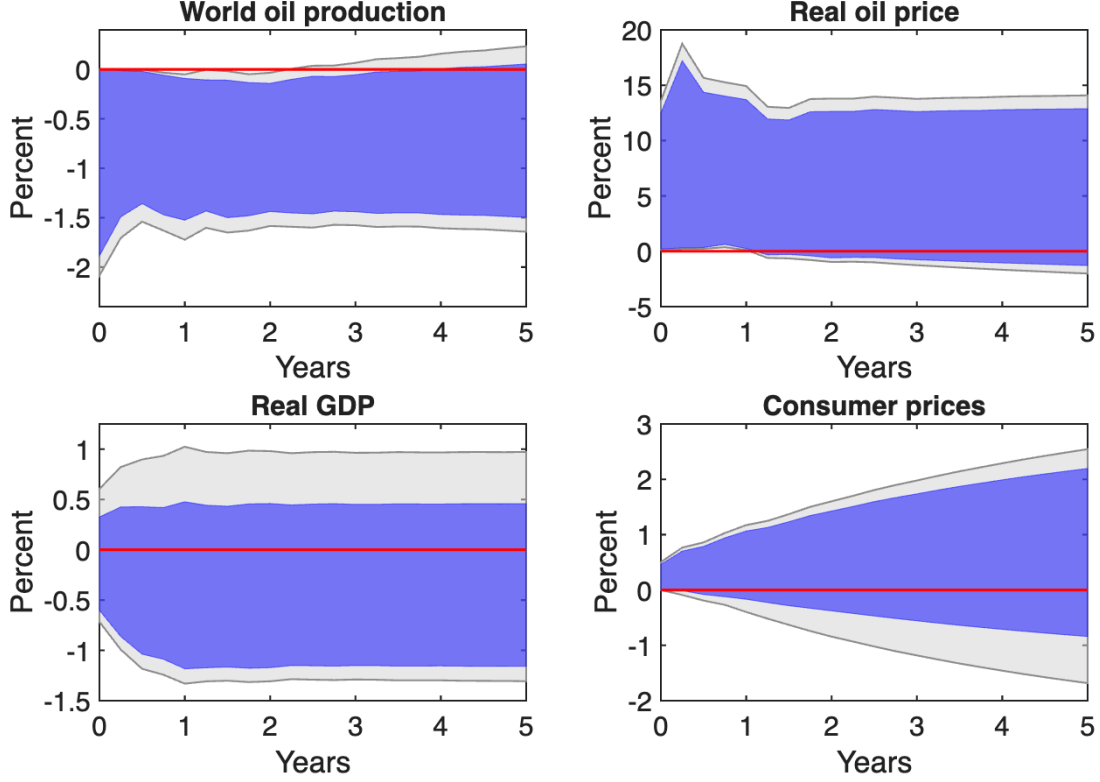


Figure E.2: Set of posterior means for an oil supply shock

Notes: The lighter shaded area is the set of posterior means using **Sign Restriction 2**. The darker blue shaded area is the set of posterior means using **Sign Restriction 2** and **Correlation Restriction 2**. The IRFs of all variables are accumulated to the log level.

sponses. The correlation restriction substantially reduces the range of these responses. These reductions support the main result that the correlation restriction helps identify oil supply shocks in a way that is consistent with the supply shock interpretation. Taken together, the monetary policy and oil supply results show that the correlation restriction adds identifying content beyond the sign restrictions and sharpens the IRFs.

F Results under alternative correlation horizons

Here, we examine whether the empirical results are sensitive to the correlation horizon. For both applications, we compare horizons of three and seven years with the five-year baseline. All other aspects of estimation and identification remain unchanged.

Figure F.1 shows the IRFs of the six variables to a contractionary monetary policy shock with **Sign Restriction 1** and **Correlation Restriction 1** for $H_c = 36, 60$, and 84 months.⁶ The overall pattern of the IRFs is similar across the three horizons. In particular, the posterior medians of output and the GDP deflator remain negative throughout

⁶ Of the 10,061 draws satisfying **Sign Restriction 1**, the correlation restriction retains 6,470, 5,787, and 5,459 draws for the three-year, five-year, and seven-year horizons, respectively.

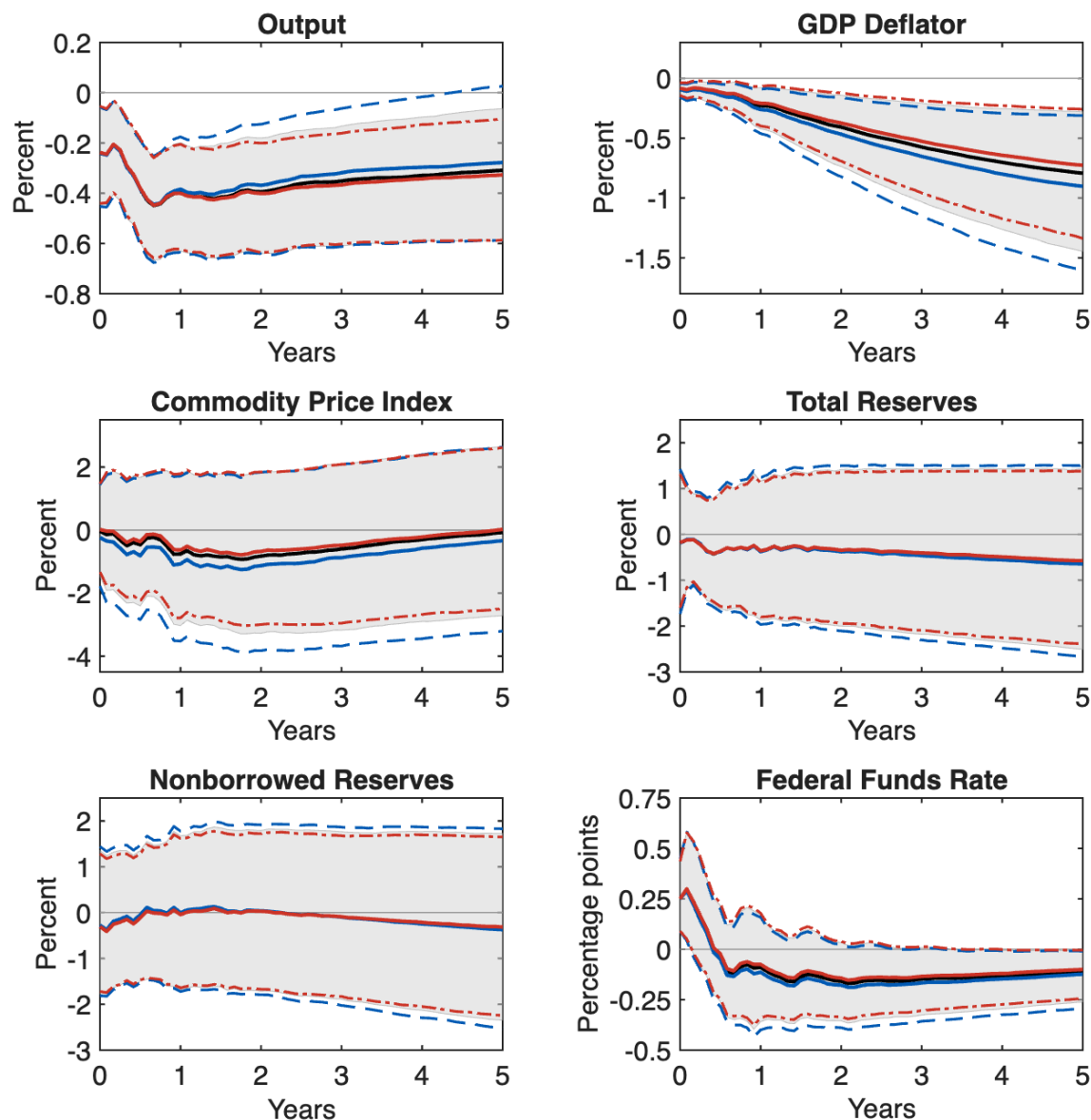


Figure F.1: IRFs to a contractionary monetary policy shock under alternative correlation horizons

Notes: All results use **Sign Restriction 1** and **Correlation Restriction 1**. The black solid lines and gray shaded areas are the median IRFs and 68% credible sets, respectively, for the five-year correlation horizon. The blue and red solid lines are the median IRFs for the three-year and seven-year horizons, respectively. The blue dashed and red dash-dotted lines mark the corresponding credible set bounds. The IRFs of all variables, except for the federal funds rate, are accumulated to the log level.

the displayed horizon. The output responses are particularly close, although the 68 percent credible set under the three-year specification includes zero at the longest response horizons. For the GDP deflator, the three-year specification implies a somewhat larger decline and a more negative lower bound at longer response horizons, but the 68 percent credible sets remain below zero throughout the displayed horizon in all three cases. These differences do not change the main conclusion that the correlation restriction helps resolve the price puzzle.

Turning to oil supply shocks, Figure F.2 compares the IRFs with **Sign Restriction 2**

and **Correlation Restriction 2** for $H_c = 12, 20$, and 28 quarters.⁷ The responses of world oil production and the real oil price remain close across the three correlation horizons. A shorter correlation horizon yields a somewhat larger median decline in output and a smaller median increase in prices. Nevertheless, the posterior medians of output remain negative and those of prices remain positive throughout the displayed horizon in all three cases. The 68% credible sets for output lie below zero from quarter 2 onward in all cases, while those for prices lie above zero initially but include zero at longer horizons. Thus, the main finding of a persistent output decline and an initial price increase is maintained across the three correlation horizons.

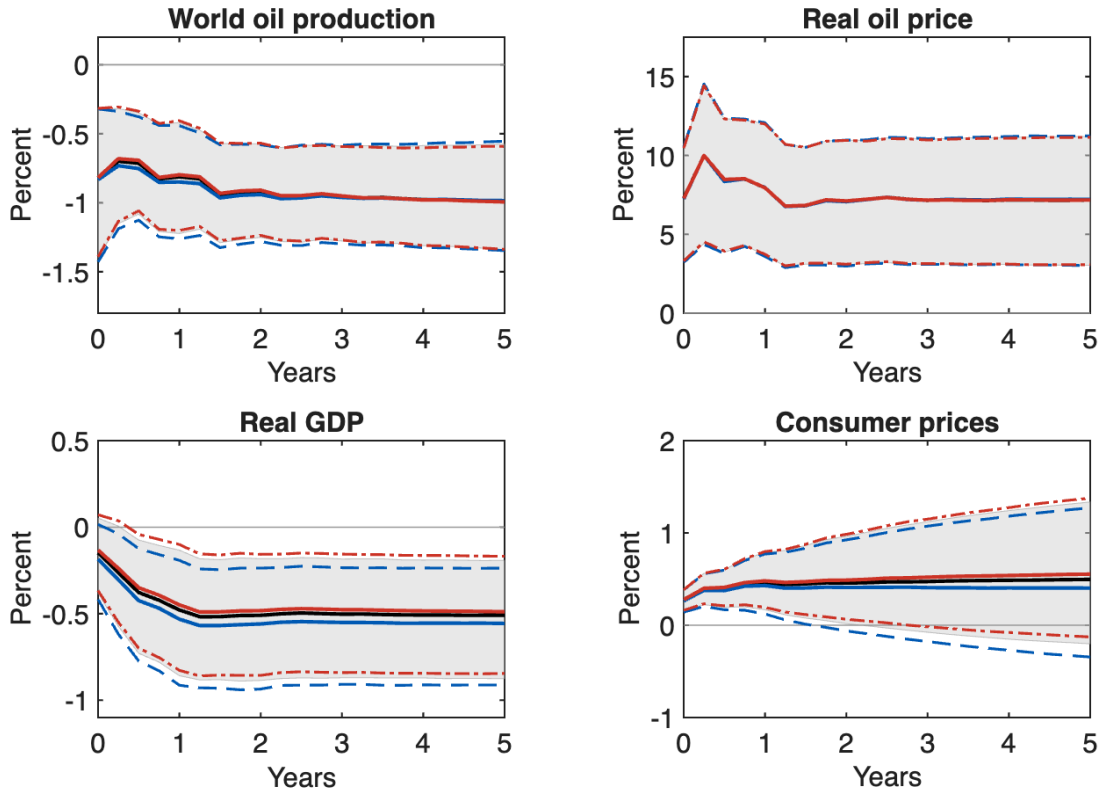


Figure F.2: IRFs to an oil supply shock under alternative correlation horizons

Notes: All results use **Sign Restriction 2** and **Correlation Restriction 2**. The black solid lines and gray shaded areas are the median IRFs and 68% credible sets, respectively, for the five-year correlation horizon. The blue and red solid lines are the median IRFs for the three-year and seven-year horizons, respectively. The blue dashed and red dash-dotted lines mark the corresponding credible set bounds. The IRFs of all variables are accumulated to the log level.

⁷ Of the 9,000 draws satisfying **Sign Restriction 2**, the correlation restriction retains 5,104, 5,028, and 4,928 draws for the three-year, five-year, and seven-year horizons, respectively.